

Final

Cluster Analysis -- Basic Concepts and Algorithms

Peerapon S.

Machine Learning (1/67)

Topics

Basic K-means algorithm

Choosing number of clusters

Insights from clustering

Clustering

Questions

What are customer behaviors?

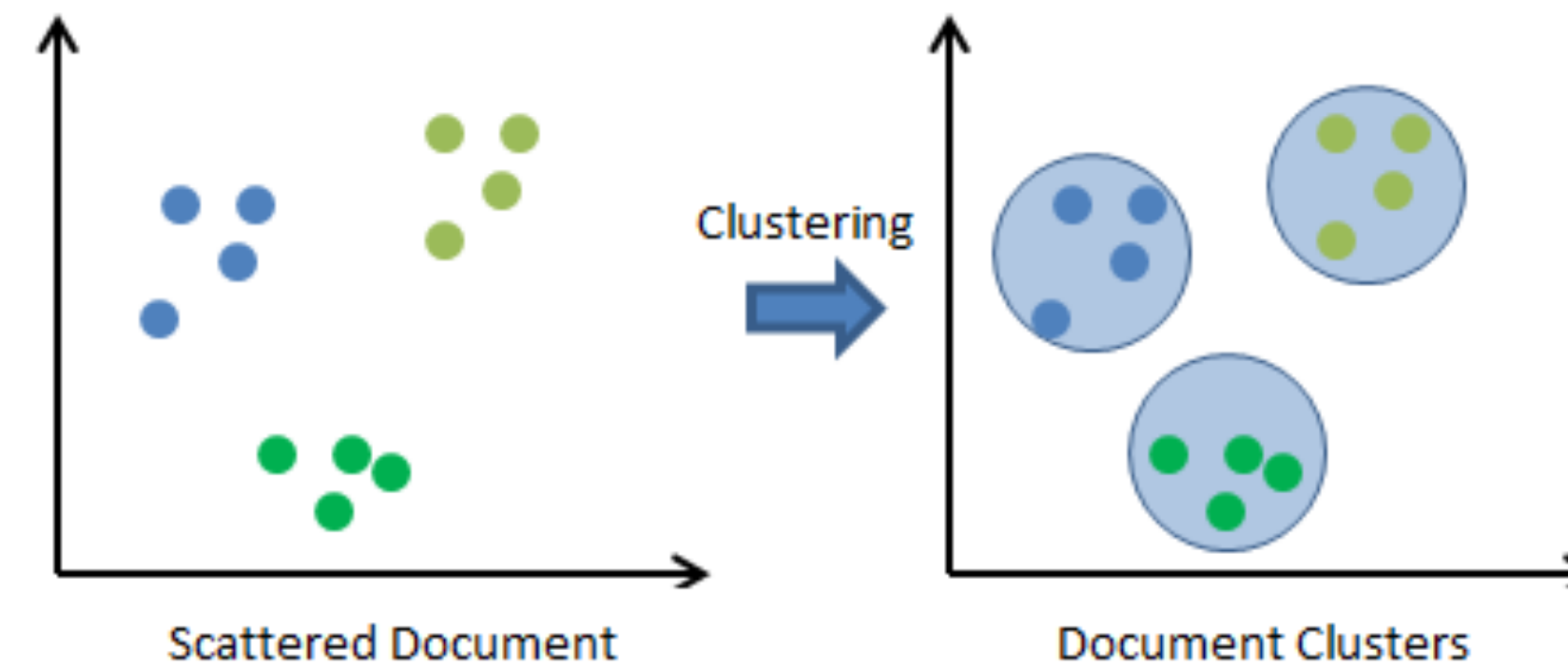
Who are our customers?

What are groups of species or syndromes, documents, etc. ?

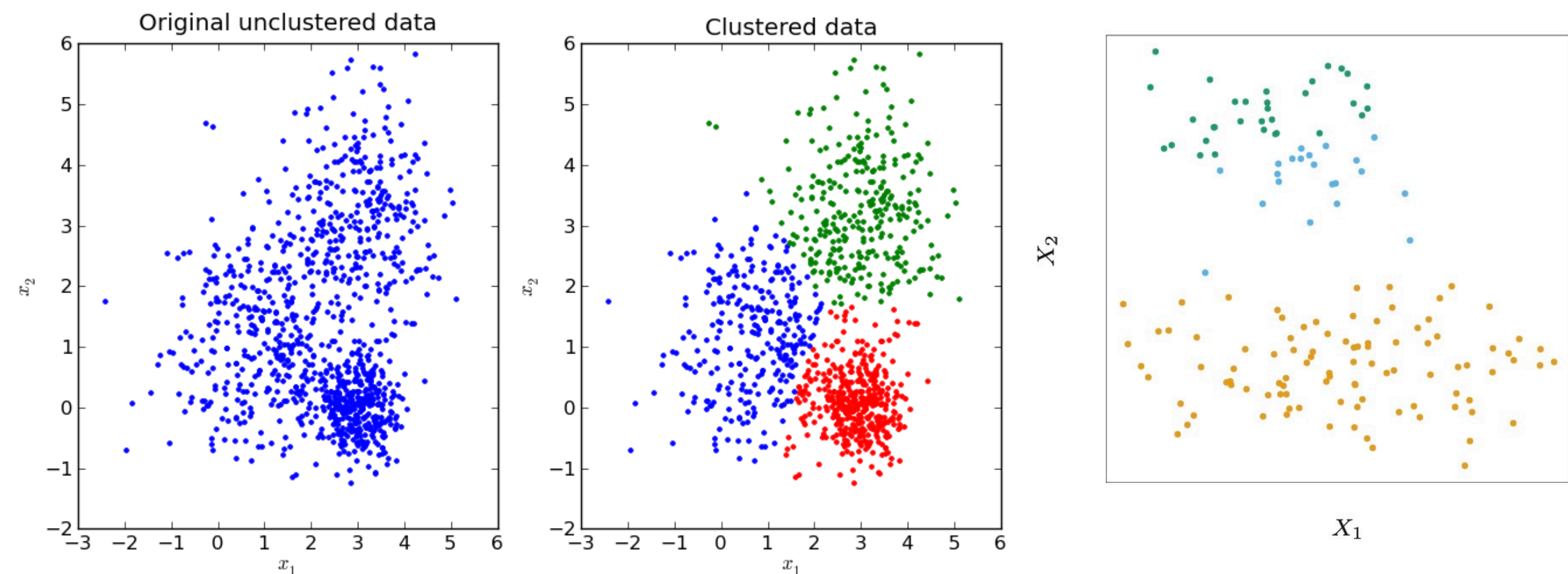
No Y data !

Goal

Find natural grouping of data



A Clustering =
A set of clusters



Centroid of a Cluster

Imaginary location representing the center of a cluster.

For a d -dimensional data set,

- ◆ The centroid of a cluster is the vector with d elements.
- ◆ Average of (scaled) numeric or (encoded) ordinal variables

In representative/prototype-based clustering, objects are assigned to their closest centroids.

	Age	Height	Weight
Person 1	45	170	64
Person 2	31	172	77
Person 3	29	159	56
Person 4	64	163	72

Centroid $\mu = \{42.25, 166, 67.25\}$

K-Means Clustering

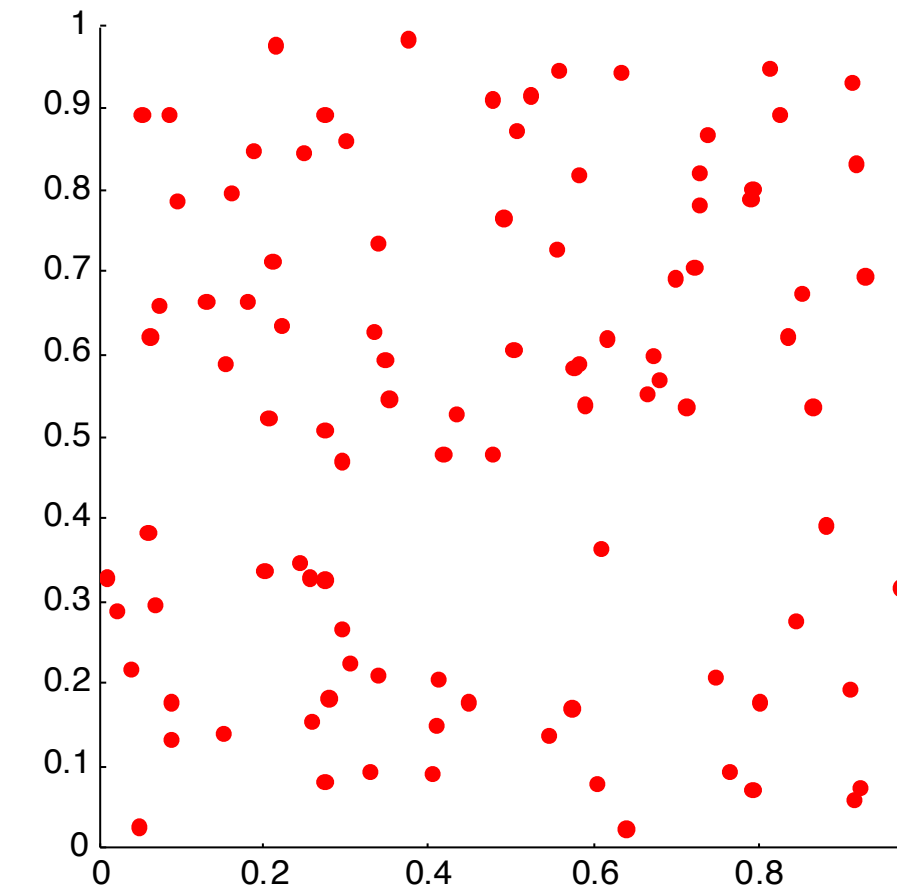
Given n data points of dimension d and a clustering $C = \{C_1, C_2, \dots, C_K\}$ with centroids μ_k , $k = 1, 2, \dots, K$.

Define the sum of squared errors of cluster as

$$\text{SSE}(\mathcal{C}) = \sum_{k=1}^K \sum_{\mathbf{x}_j \in C_k} \|\mathbf{x}_j - \mu_k\|^2$$

Find the clustering $C^* = \{C_1, C_2, \dots, C_K\}$ that minimizes SSE: $\mathcal{C}^* = \arg \min_{\mathcal{C}} \text{SSE}(\mathcal{C})$

Optimal clustering requires exhaustive search !

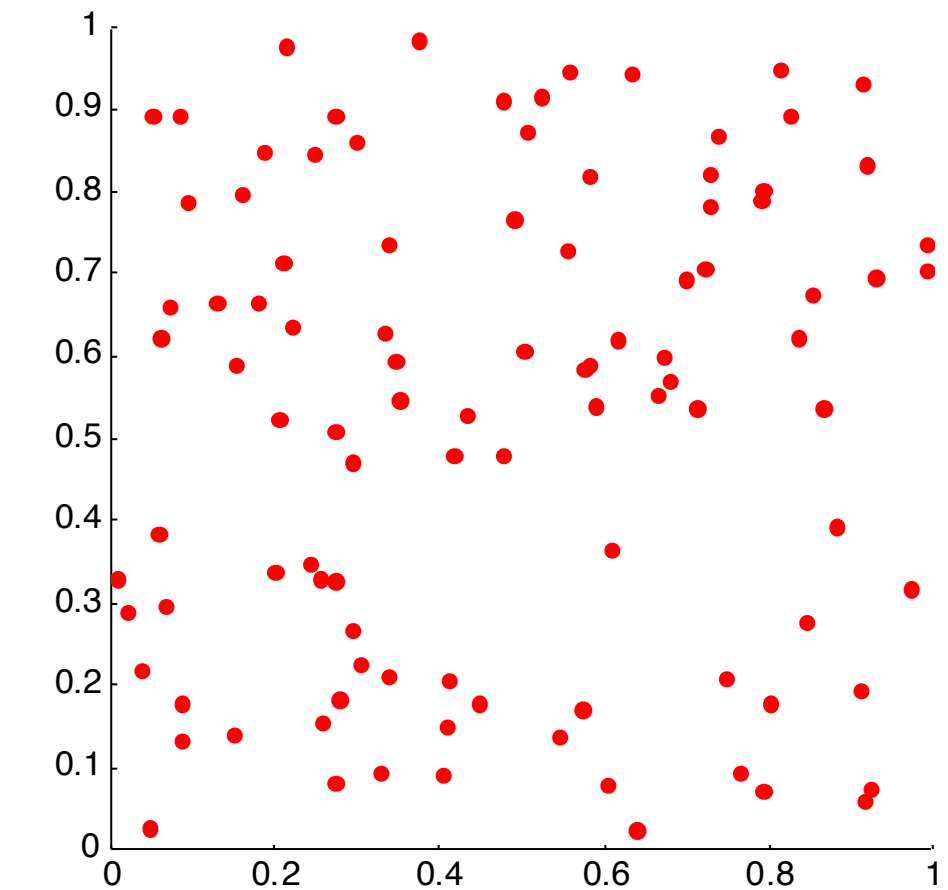


Greedy iterative approach is used to find a clustering that minimizes the SSE objective.

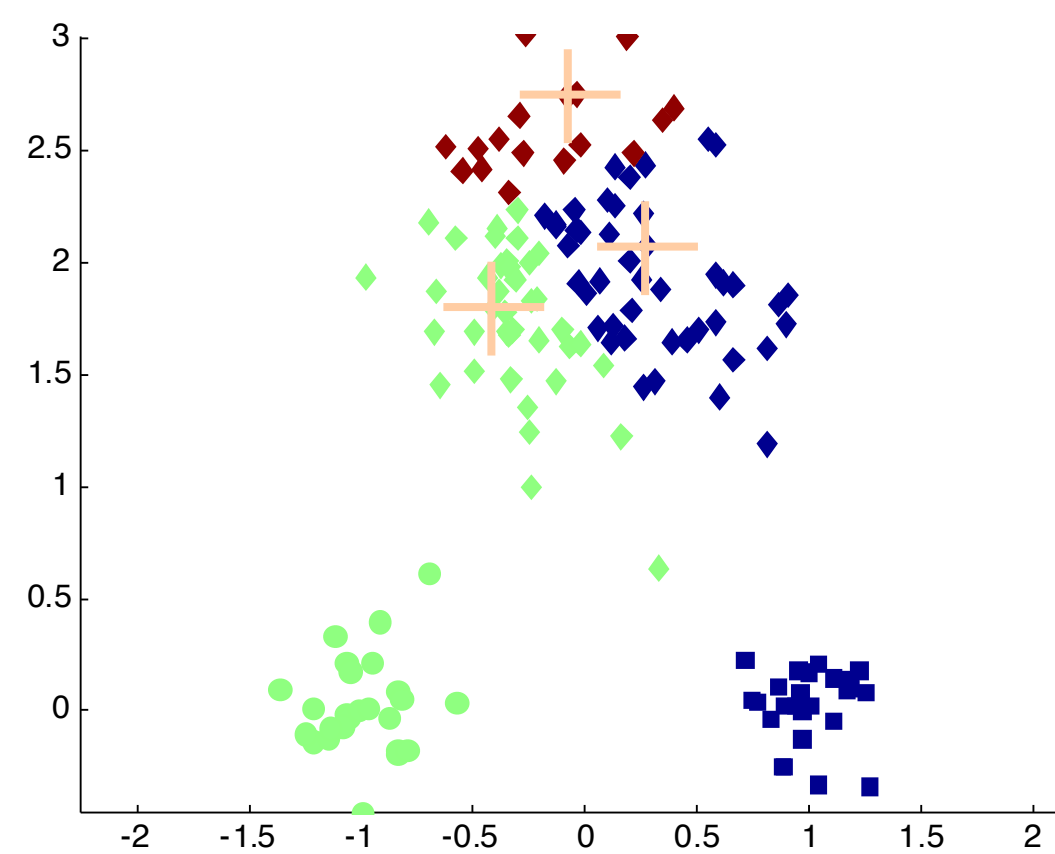
Provide good-enough solutions but not guarantee optimality.

Lloyd's K-means algorithm

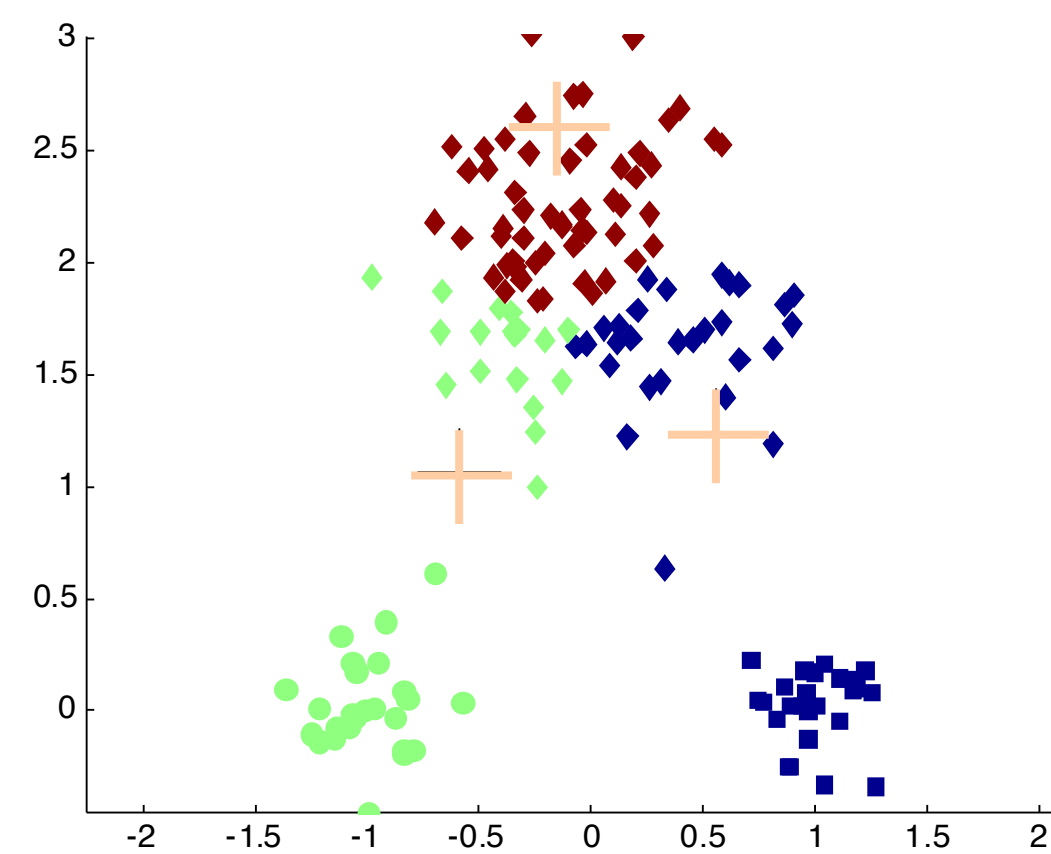
- 1: Select K points as the initial centroids.
 - 2: **repeat**
 - 3: Form K clusters by assigning all points to the closest centroid.
 - 4: Recompute the centroid of each cluster.
 - 5: **until** The centroids don't change
-



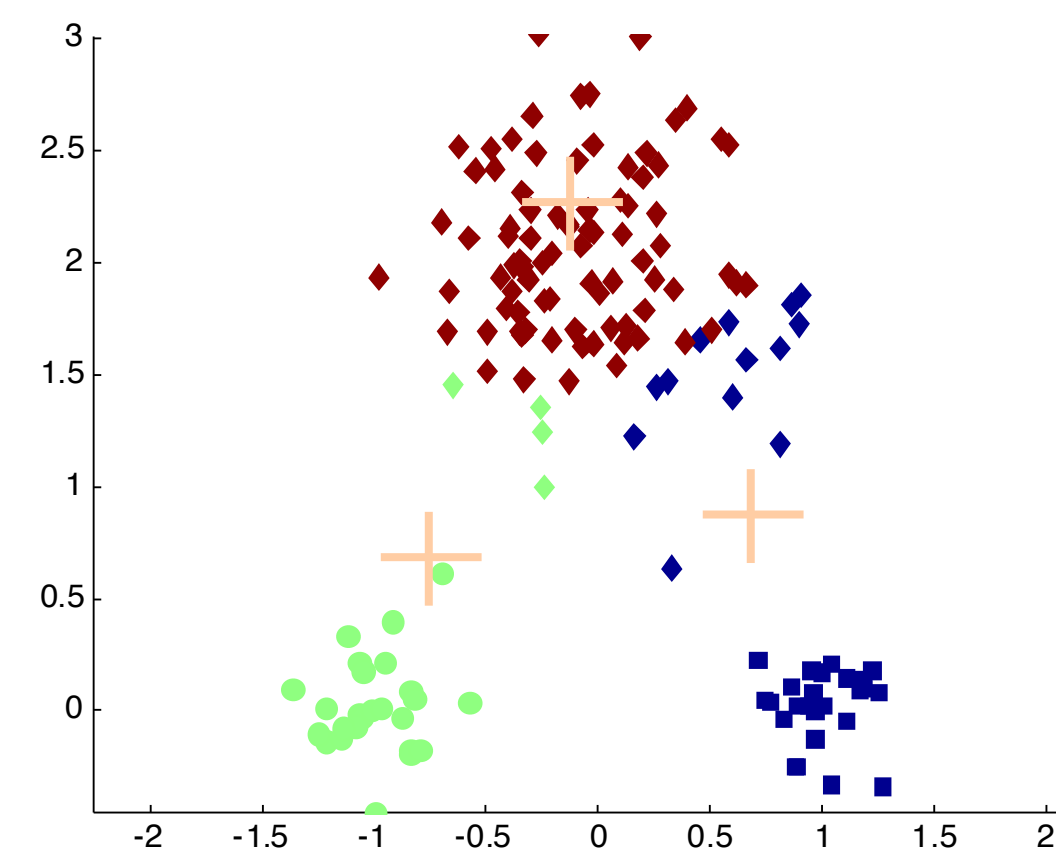
$$K = 3$$



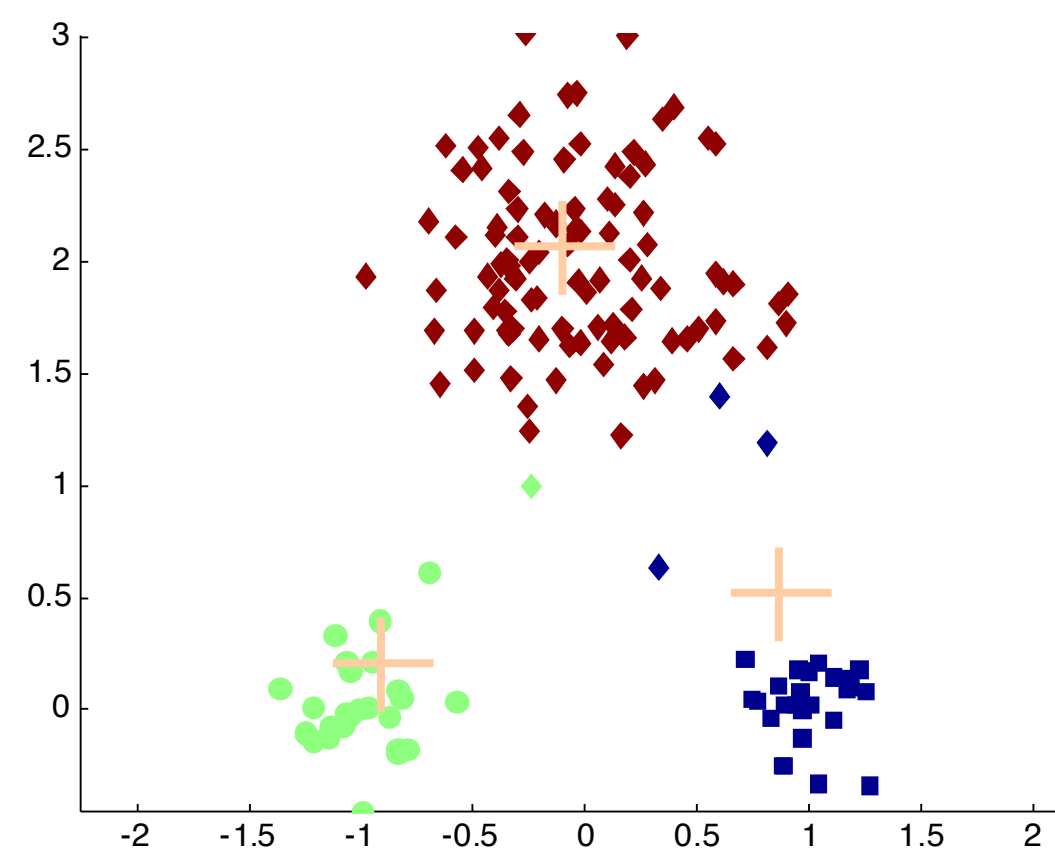
Iteration 1



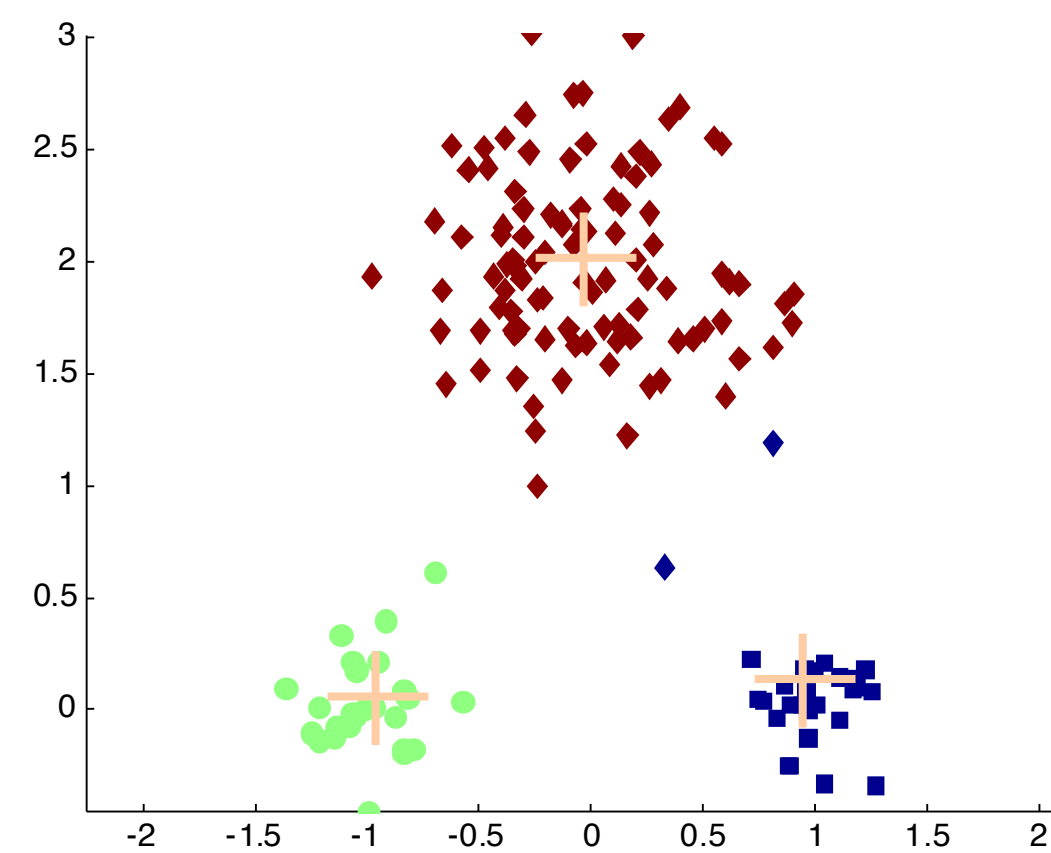
Iteration 2



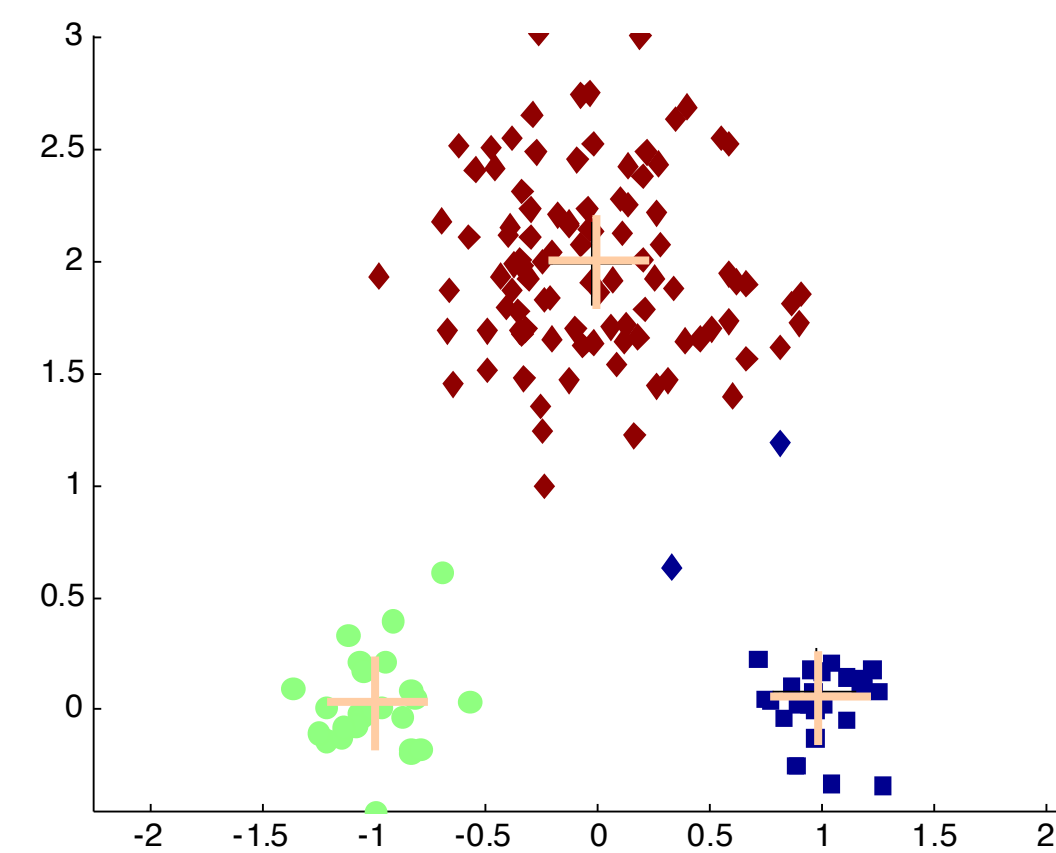
Iteration 3



Iteration 4

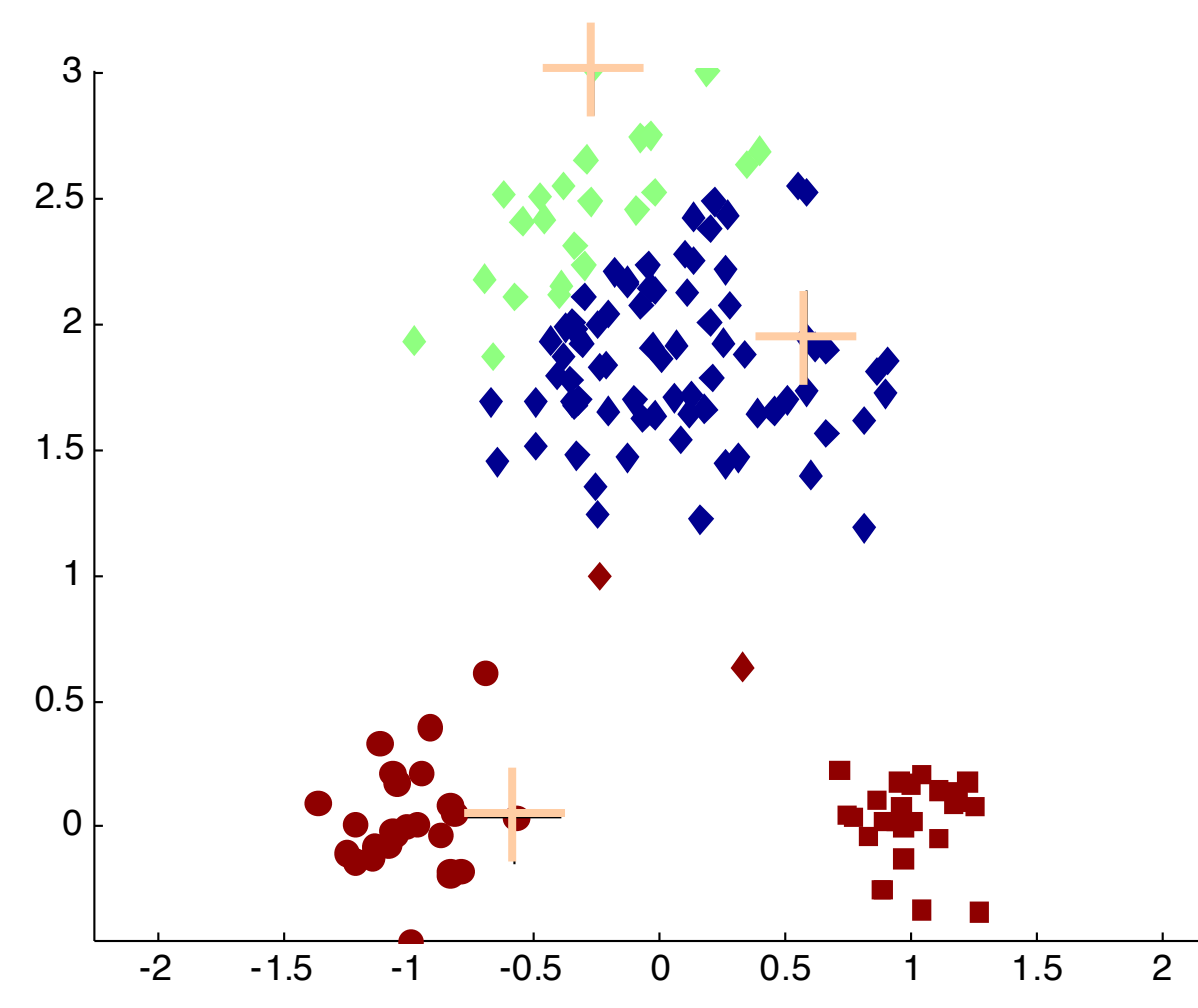


Iteration 5

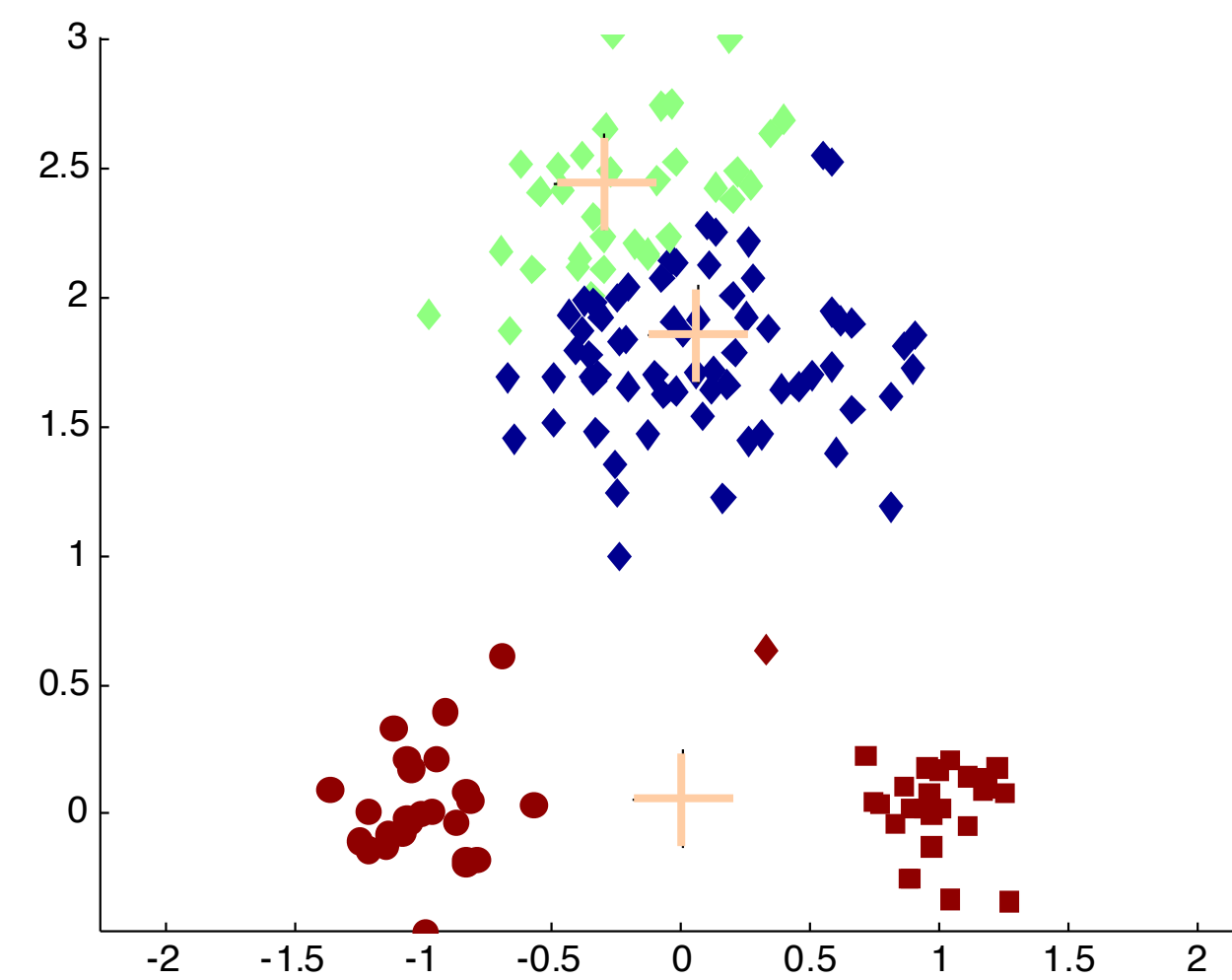


Iteration 6

Importance of Initial Conditions



Initialization



Termination



```
from sklearn.datasets import make_blobs

X, _ = make_blobs(n_samples=800, centers=4, random_state=42)
plt.scatter(X[:, 0], X[:, 1]);

from sklearn.cluster import KMeans
# Number of clusters
K = 6

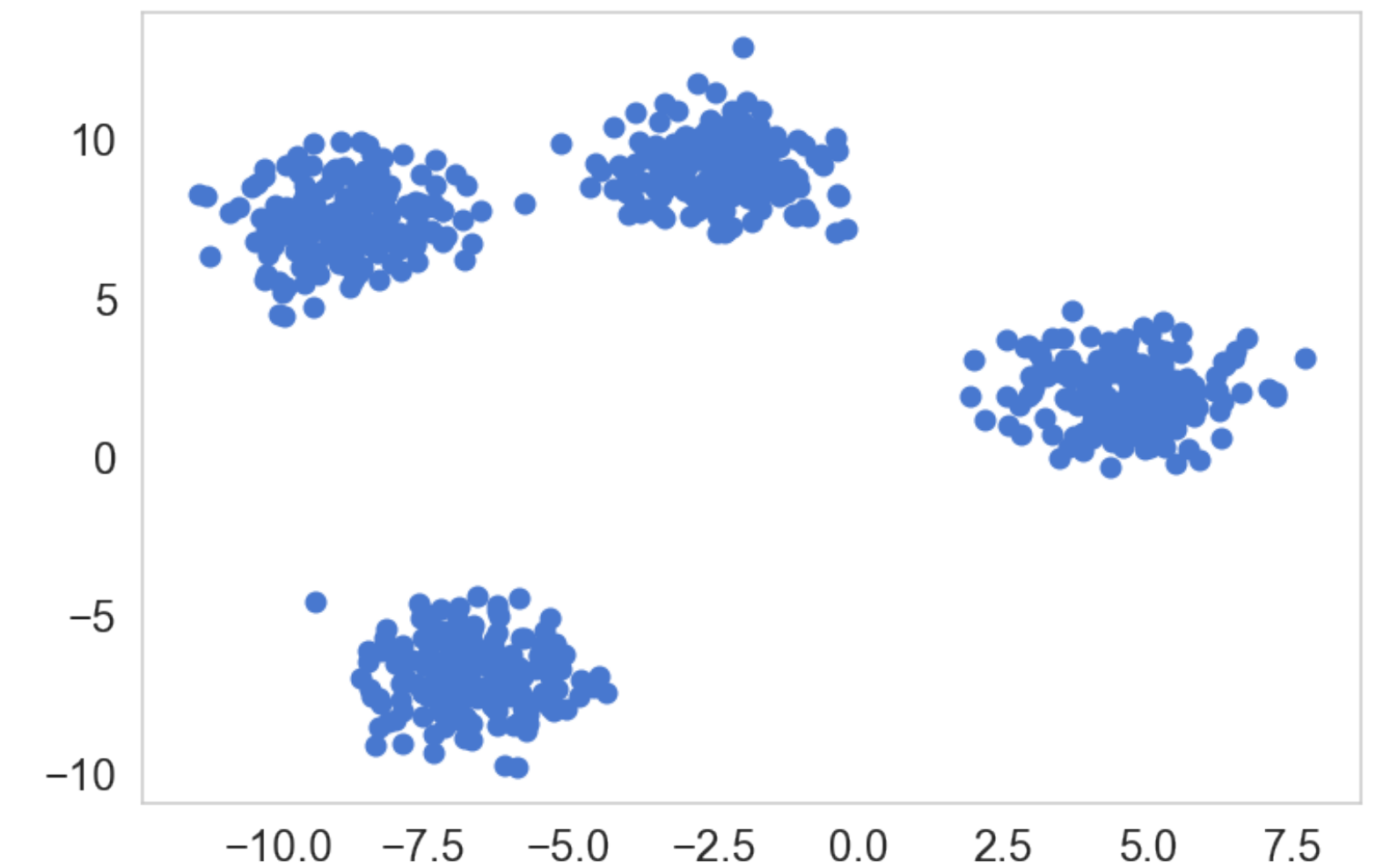
# Fit the input data
kmeans = KMeans(n_clusters=K, n_init=10).fit(X)

# Get the cluster labels
labels = kmeans.predict(X) # or use kmeans.labels

# Get sum of squares distance of all points
sse = kmeans.inertia_

# Centroid values
centroids = kmeans.cluster_centers_

np.set_printoptions(precision=3)
print(f"Sum of squared errors : {sse:.3f}")
print(f"\nCentroids : \n {centroids}")
```



Sum of squared errors : 1273.890

Centroids :

[[-8.262	8.17]
[-6.682	-6.81]
[5.041	1.4]
[-2.501	9.053]
[3.96	2.726]
[-9.276	6.737]]

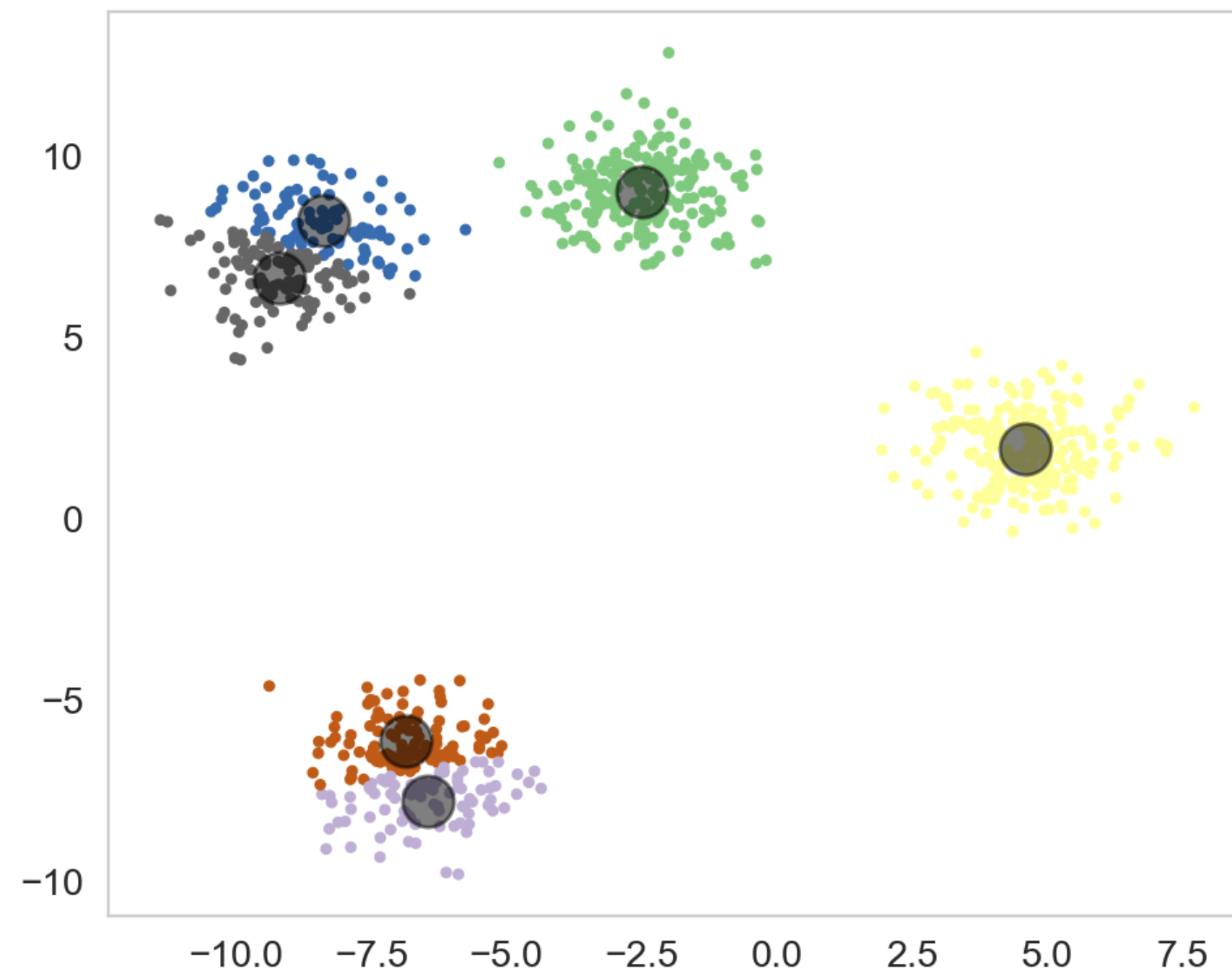


```
import matplotlib.pyplot as plt
```

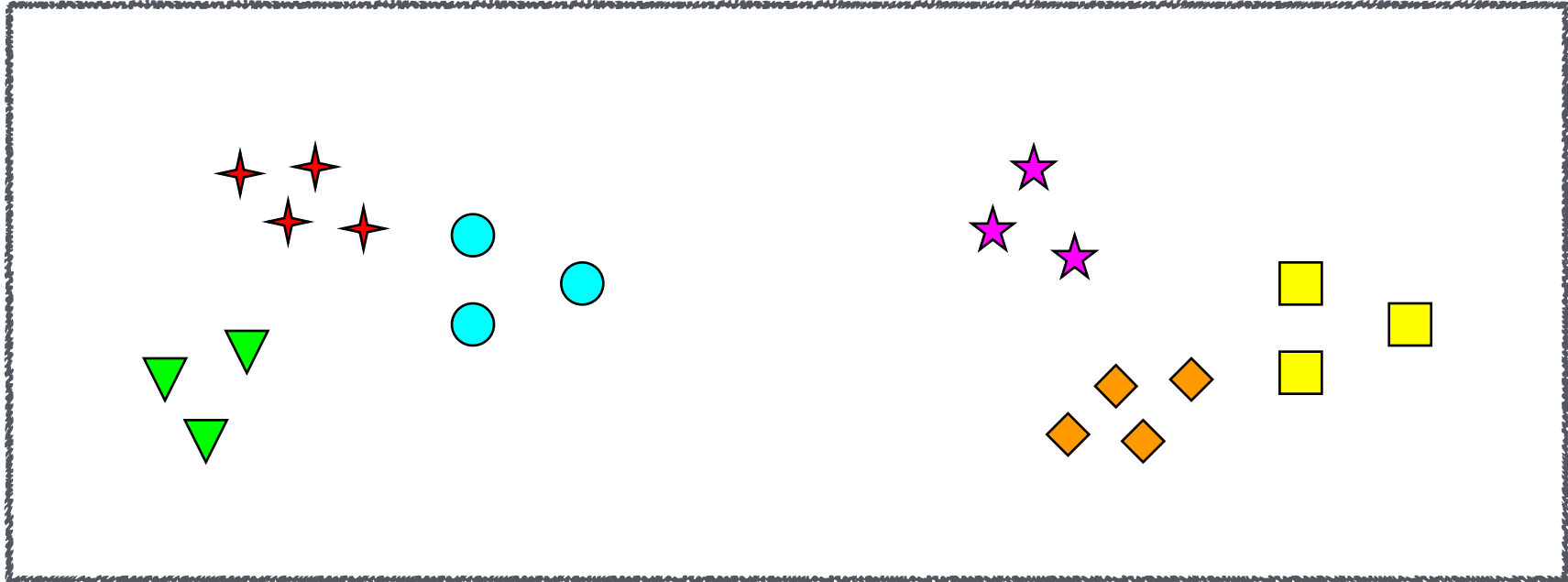
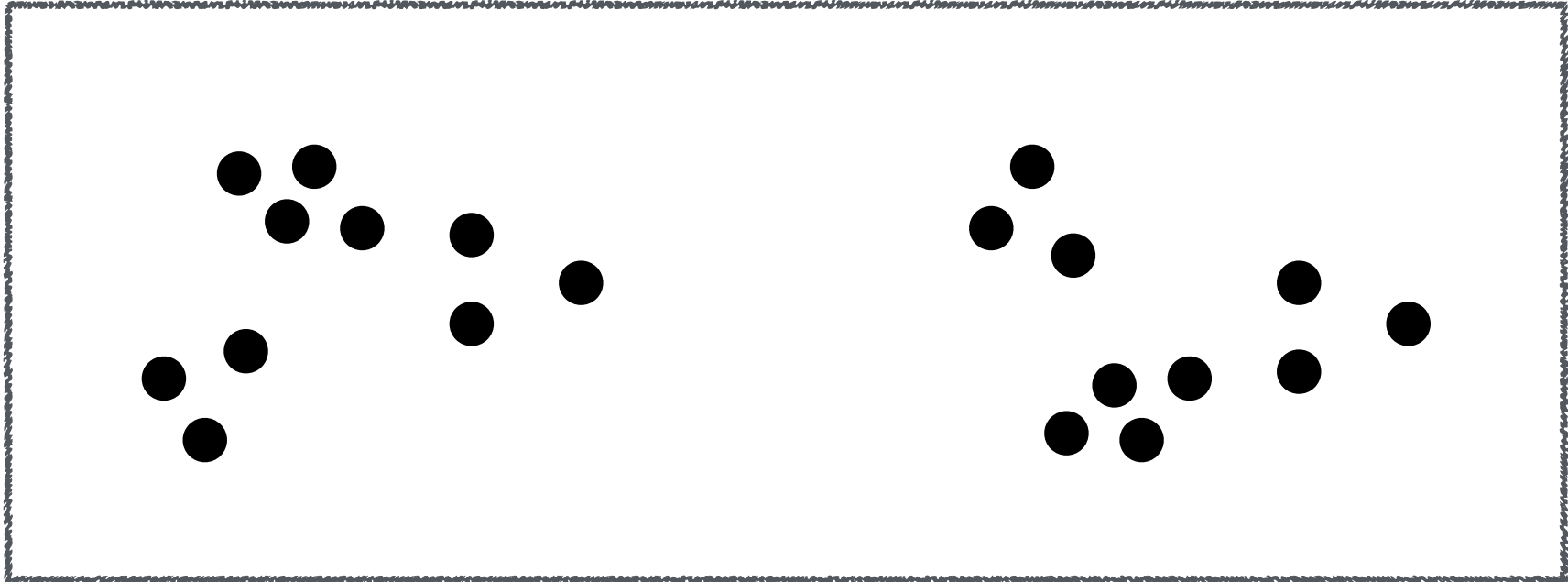
```
plt.scatter(X[:, 0], X[:, 1], c=labels, s=5, cmap='Accent')
```

```
centers = kmeans.cluster_centers_
```

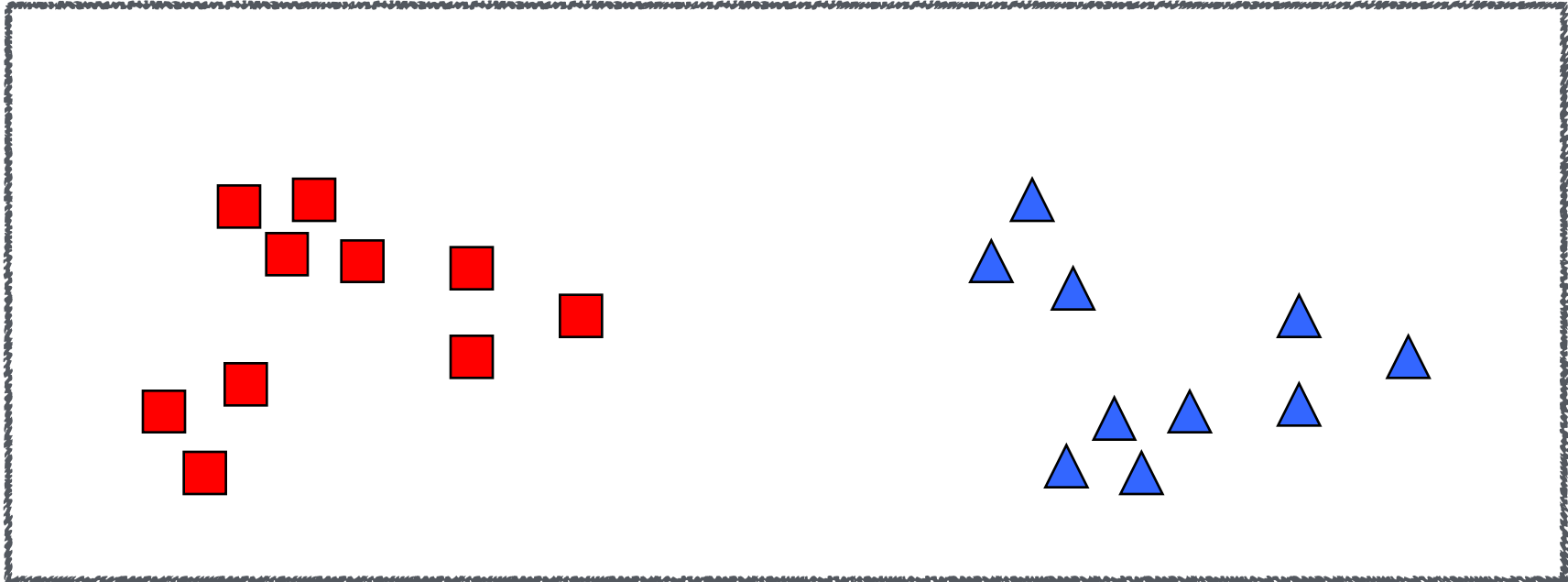
```
plt.scatter(centers[:, 0], centers[:, 1], c='black', s=200, alpha=0.5);
```



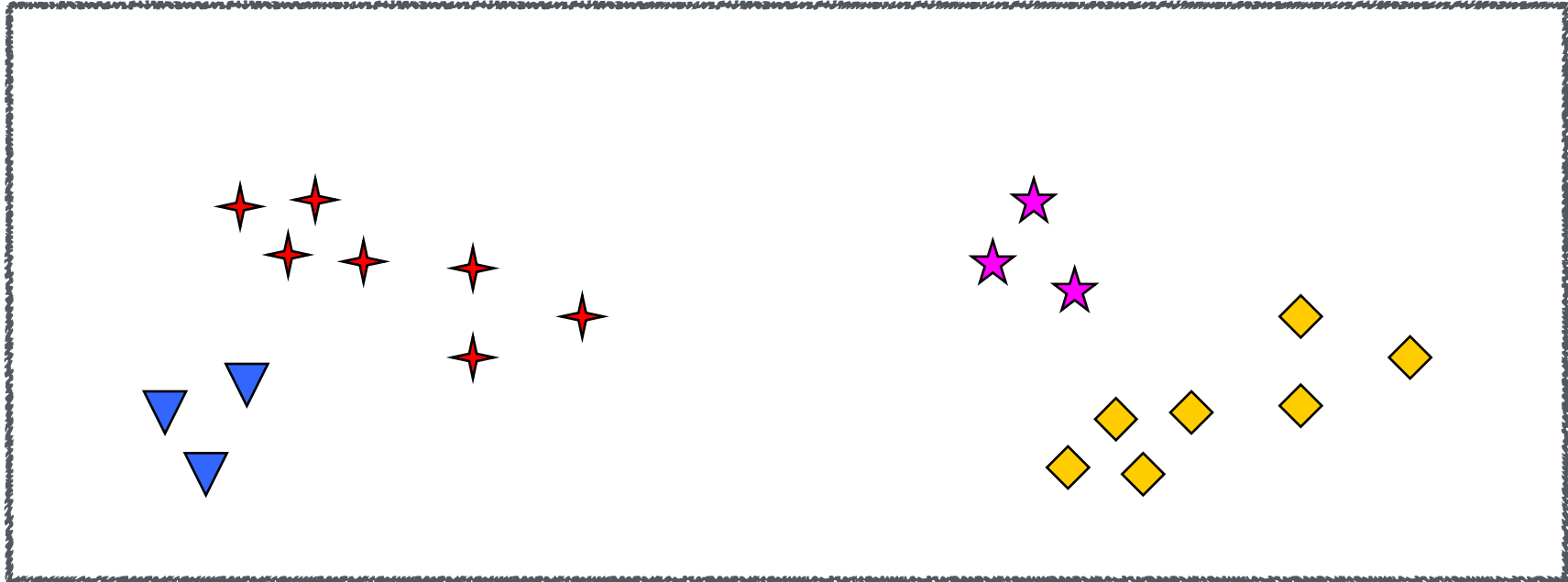
How many clusters ?



Six Clusters



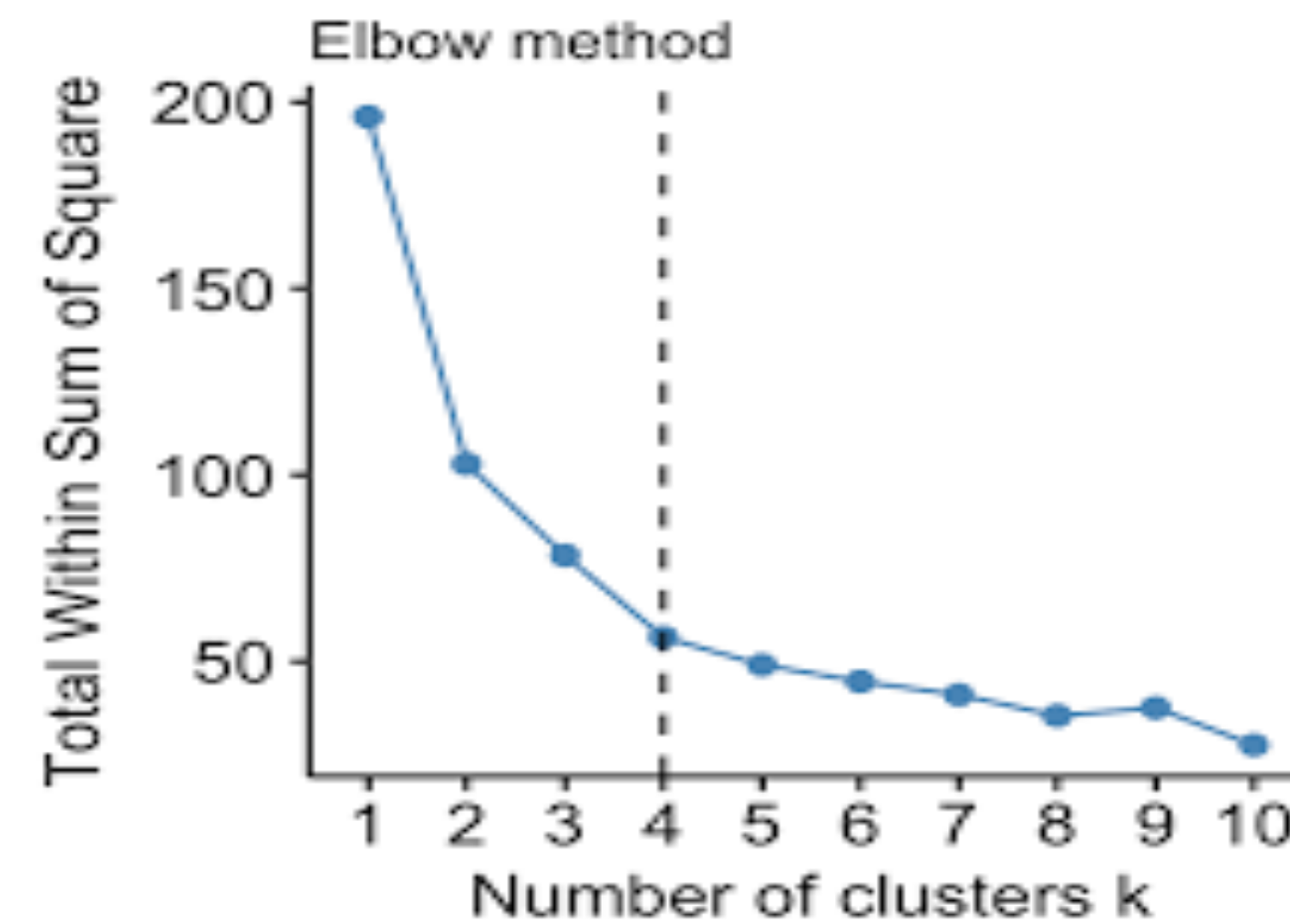
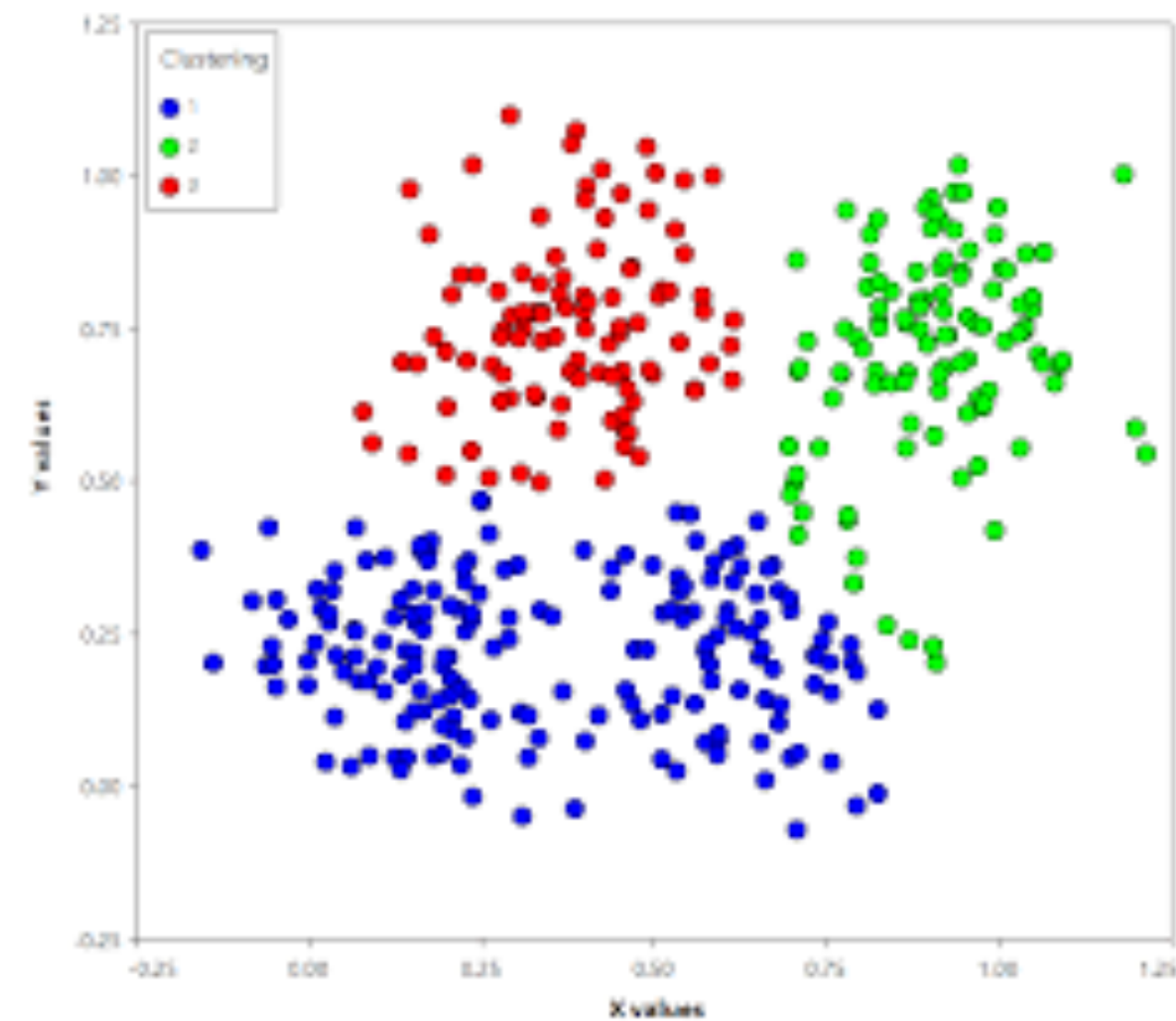
Two Clusters



Four Clusters

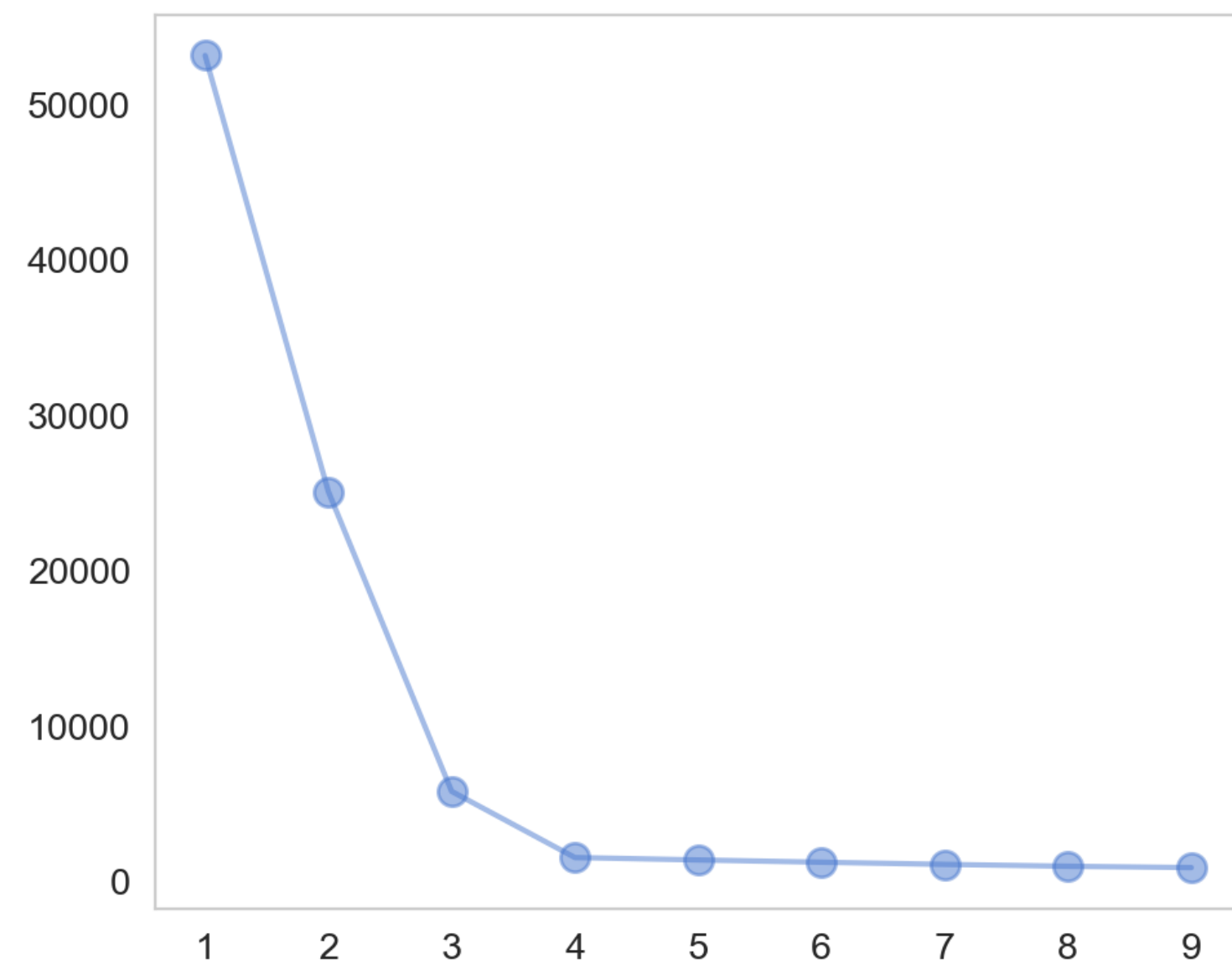
Choosing the Right K : **Elbow Method**

Look at the smallest possible K where SSE no longer has significant improvement, i.e., the *knee point*.

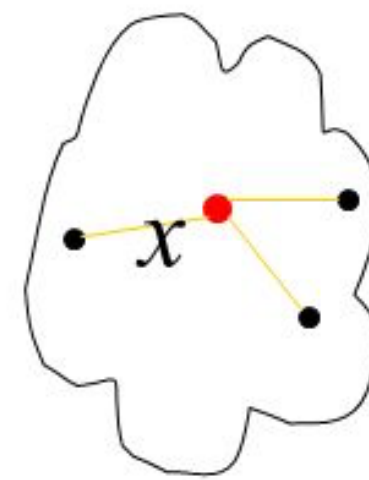




```
sse = {}  
for k in range(1, 10):  
    kmeans = KMeans(n_clusters=k).fit(X)  
    sse[k] = kmeans.inertia_  
  
_,ax=plt.subplots(figsize=(5,4))  
ax.plot(list(sse.keys()), list(sse.values()), marker='o', alpha=0.5, ms=8);  
plt.tight_layout();
```

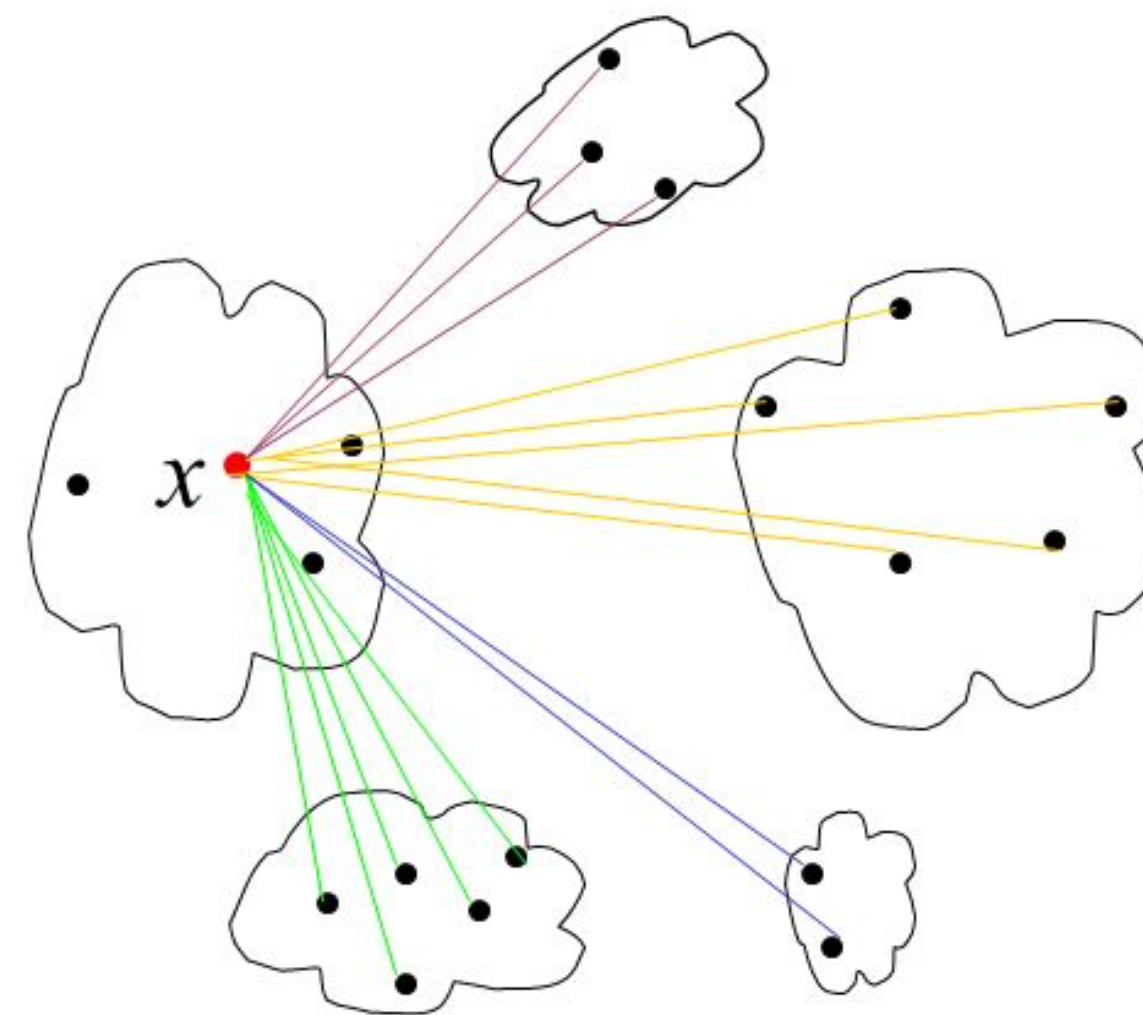


Silhouette Coefficient



Cohesion

a_i : Average distance to points within cluster



Separation

b_i : Minimum average distance to points in a different cluster

$$s_i = \frac{b_i - a_i}{\max(a_i, b_i)} = \begin{cases} 1 - a_i/b_i & \text{if } a_i < b_i \\ 0 & \text{if } a_i = b_i \\ b_i/a_i - 1 & \text{if } a_i > b_i \end{cases}$$

Silhouette coefficient (SC) = $n^{-1} \sum_{i=1}^n s_i$

$[-1, 1]$
↓ ↓
negatif positif

Calinski Harabasz Index

Also known as *Variance Ratio Criterion* (higher is better).

Ratio of sum of *between-clusters dispersion* and *within-clusters dispersion* for all clusters.

Define

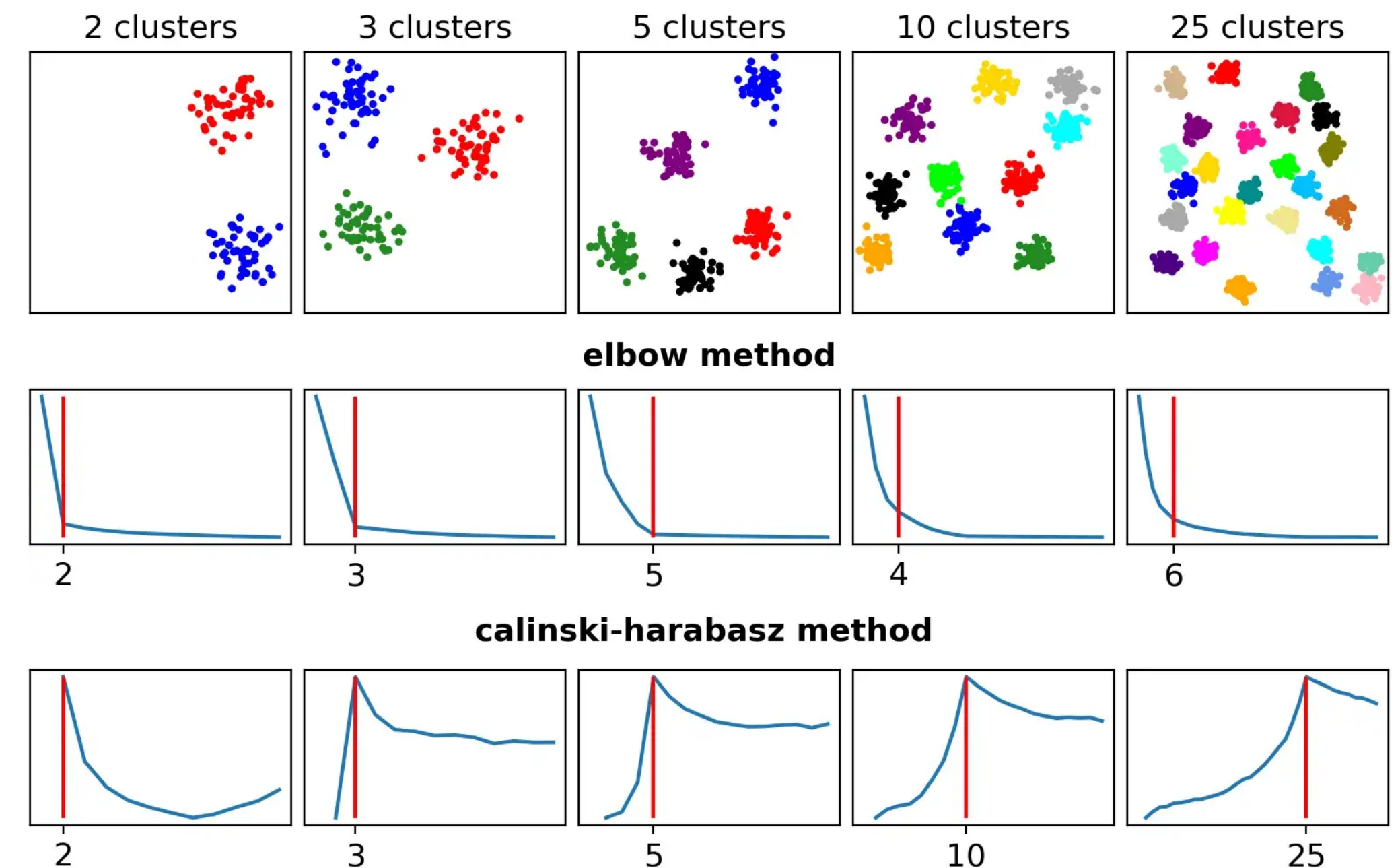
- ◆ N data points with global center $\mathbf{c}_G$.
- ◆ K clusters, each with N_q data points and center $\mathbf{c}_q$ in cluster q .

The Calinski Harabasz index is given by

$$S = \frac{B_K / (K - 1)}{W_K / (N - K)}$$

$$B_K = \sum_{q=1}^K N_q \|\mathbf{c}_q - \mathbf{c}_G\|^2 \quad \text{Between-cluster dispersion}$$

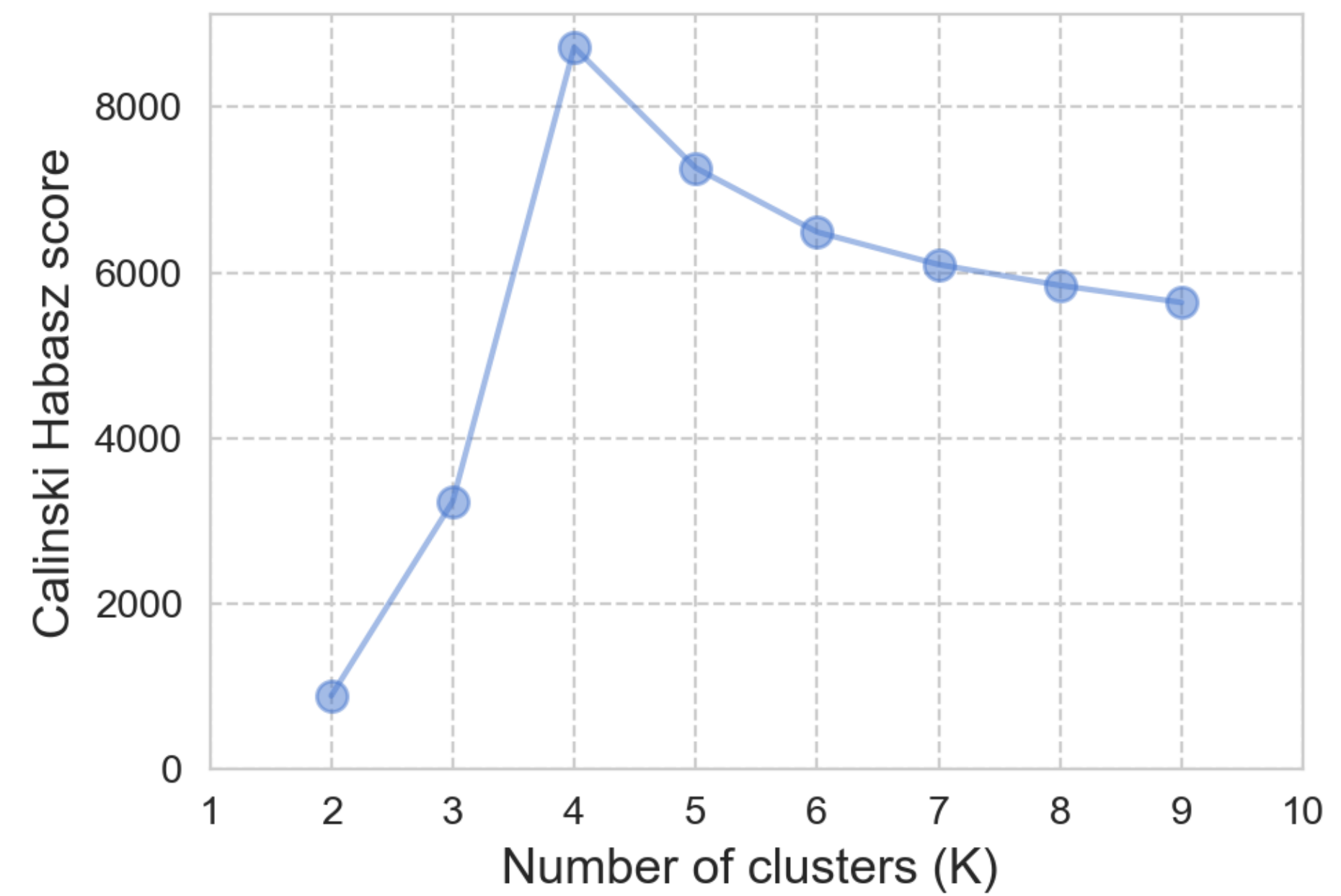
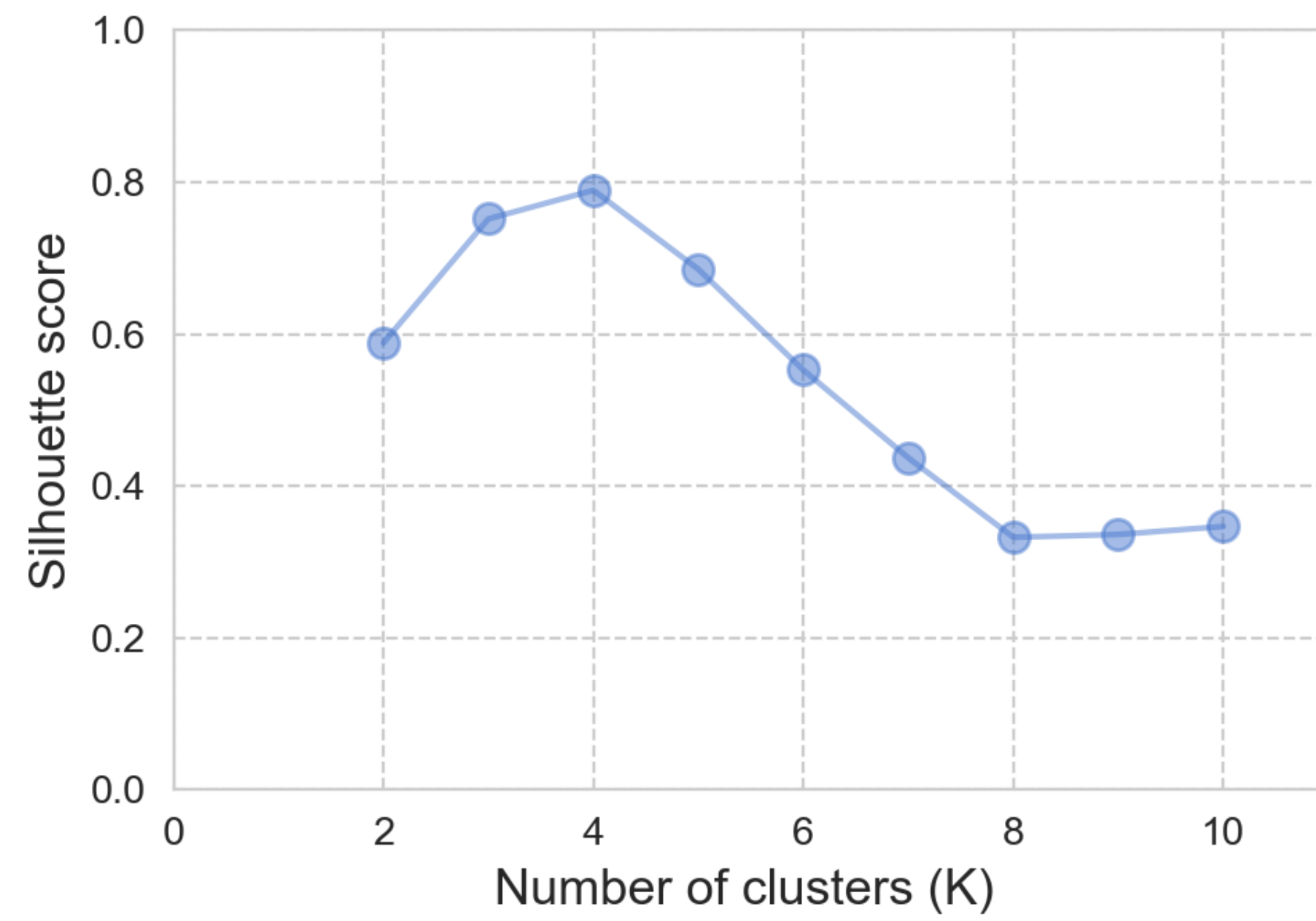
$$W_K = \sum_{q=1}^K \sum_{\mathbf{x} \in C_q} \|\mathbf{x} - \mathbf{c}_q\|^2 \quad \text{Within-cluster dispersion}$$





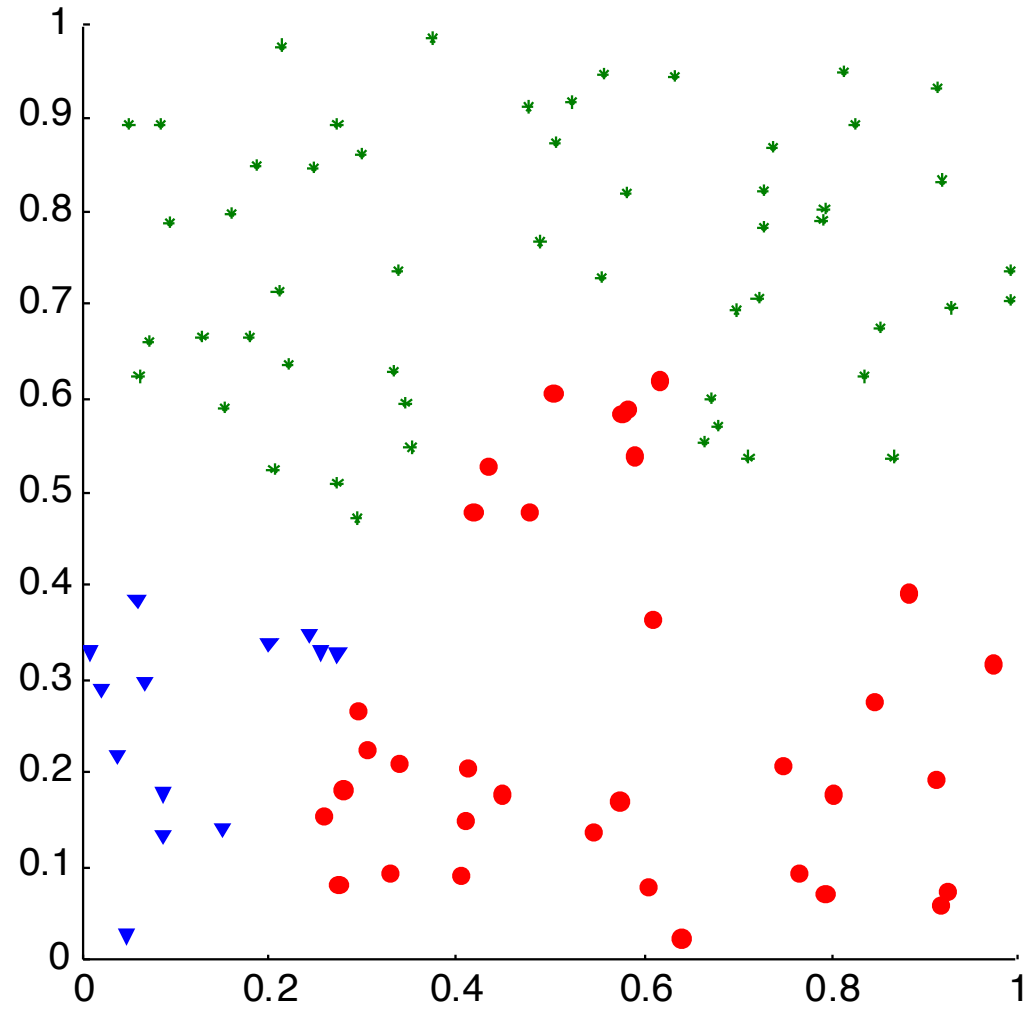
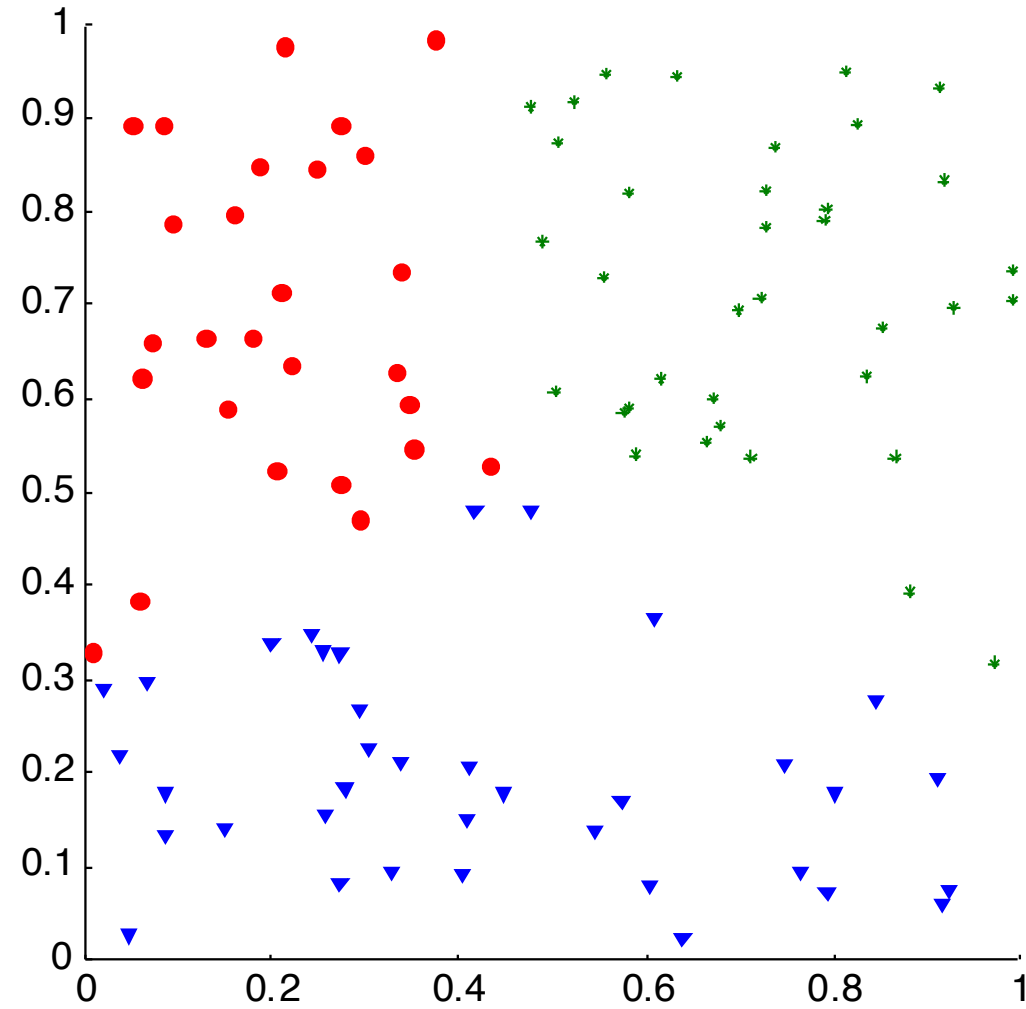
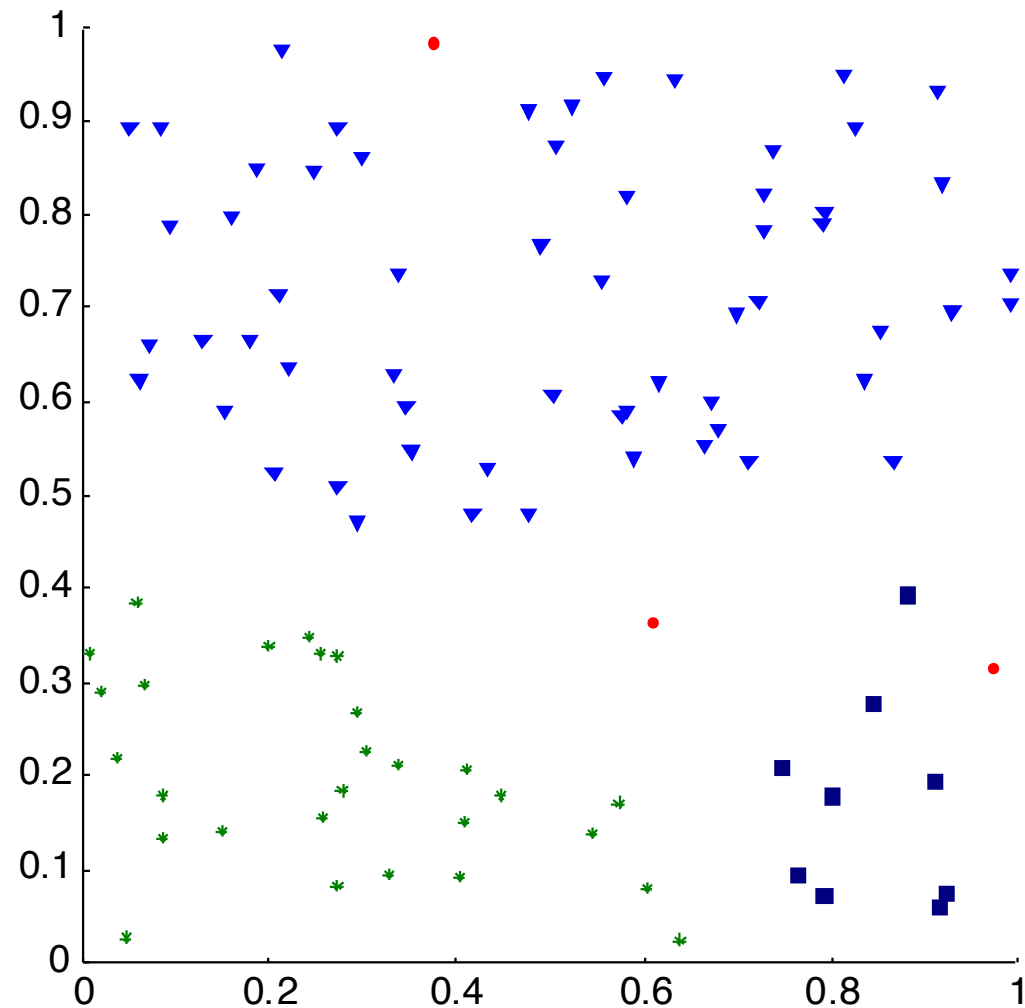
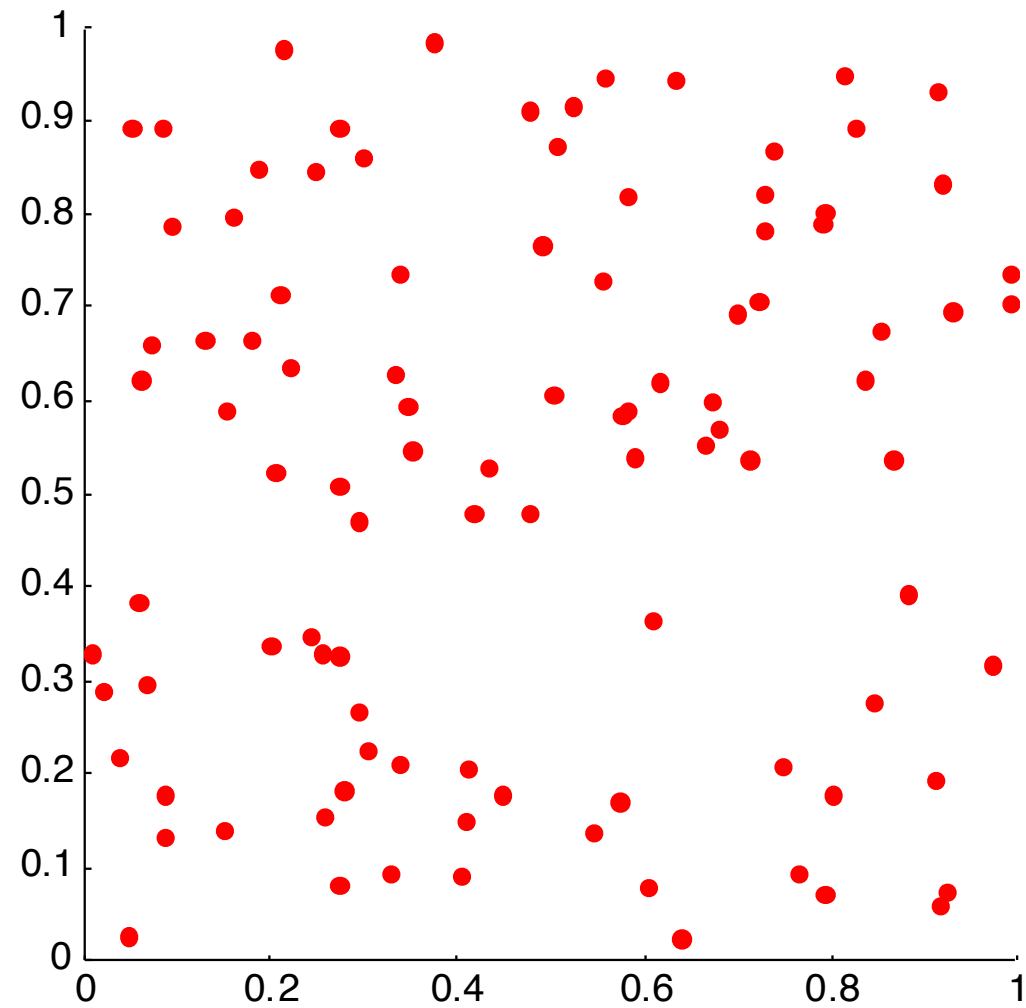
```
from sklearn.metrics import silhouette_score  
silhouette_score(X, labels)
```

```
from sklearn.metrics import calinski_harabasz_score  
calinski_harabasz_score(X, labels)
```



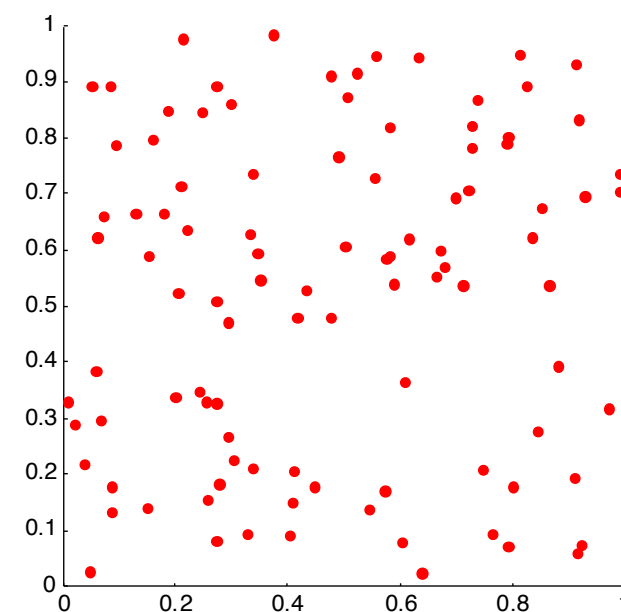
Clusters found in Random Data

Random
Points



Hopkins Statistics

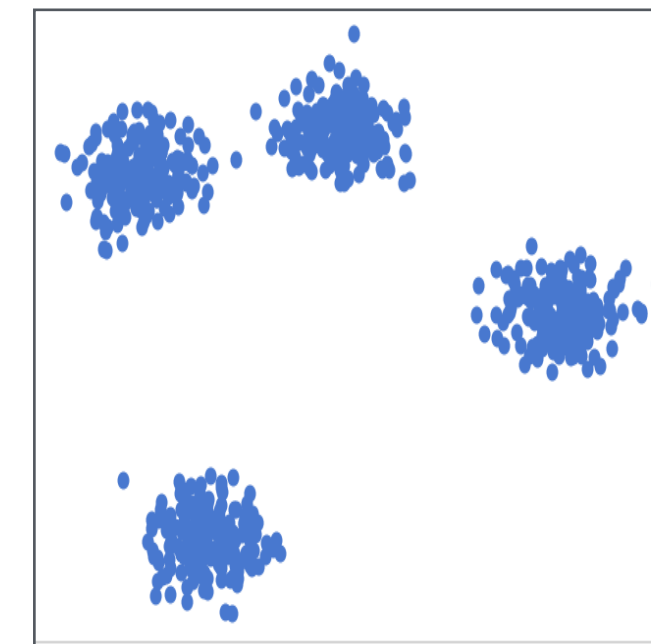
Random data



Sample p points

For each point i ,
find distance u_i to its nearest
neighbor in actual data

Actual data



Sample p points

For each point i ,
find distance w_i to its nearest
neighbor in actual data

$$H = \frac{\sum_{i=1}^p w_i}{\sum_{i=1}^p u_i + \sum_{i=1}^p w_i}$$

1 $\rightarrow$ δ pattern
0.5 $\rightarrow$ random data

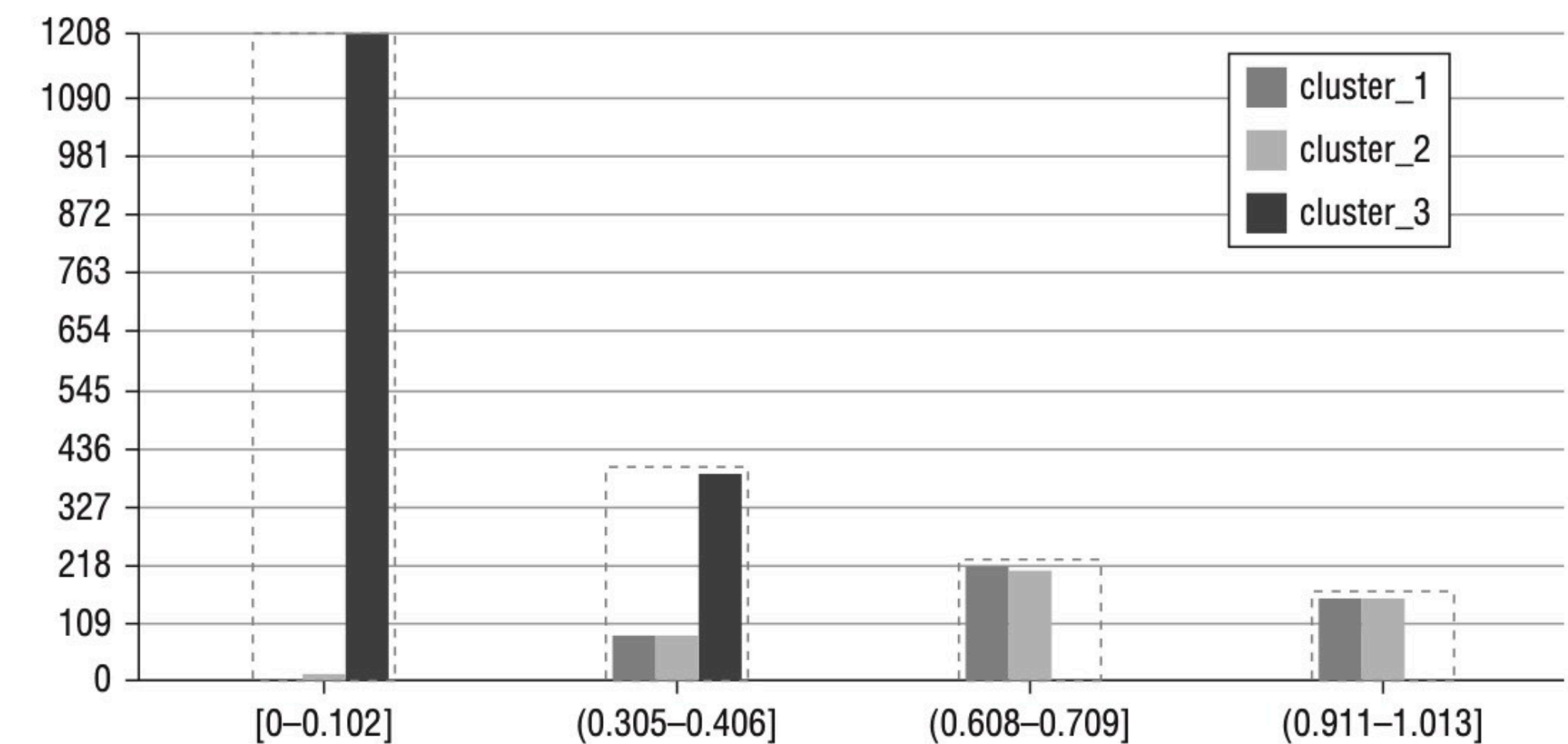
Taking Insights from Clustering

Count of records in each cluster

Summary statistics of (descaled) variables in each cluster (Mean vectors = Centroids)

ANOVA to find which variables have significant difference in means among clusters.

VARIABLE	CLUSTER 1	CLUSTER 2	CLUSTER 3	OVERALL
# Records in Cluster	8,538	8,511	30,656	47,705
LASTDATE	0.319	0.304	0.179	0.226
FISTDATE	0.886	0.885	0.908	0.900
RFA_2F	0.711	0.716	0.074	0.303
D_RFA_2A	0.382	0.390	0.300	0.331
E_RFA_2A	0.499	0.500	0.331	0.391
F_RFA_2A	0.369	0.366	0.568	0.496
DOMAIN3	0.449	0.300	0.368	0.370
DOMAIN2	0.300	0.700	0.489	0.493
DOMAIN1	0.515	0.300	0.427	0.420
NGIFTALL_log10	0.384	0.385	0.233	0.287
LASTGIFT_log10	0.348	0.343	0.430	0.400



Identifying Key Variables in Forming Clusters

May not be feasible to examine graphs/tables of every variable in every cluster.

Assign cluster labels to data and build a decision tree to find feature importance.

CLUSTER LABEL	RULE	NUMBER OF RECORDS MATCHING RULE	NUMBER OF RECORDS WITH CLUSTER LABEL IN RULE	TREE ACCURACY
Cluster 1, Rule 1	DOMAIN2 = 0 and RFA_2F > 2	6,807	6,807	100%
Cluster 1, Rule 2	DOMAIN2 = 0 and RFA_2F = 2 and E_RFA_2A = 1	1,262	1,262	100%
Cluster 2, Rule 1	DOMAIN2 = 1 and RFA_2F > 2	6,751	6,751	100%
Cluster 2, Rule 2	DOMAIN2 = 1 and RFA_2F = 2 and E_RFA_2A = 1	1,304	1,304	100%
Cluster 3, Rule 1	RFA_2F = 1 and E_RFA_2A = 0	21,415	21,415	100%
Cluster 3, Rule 2	RFA_2F = 1 and E_RFA_2A = 1	2,432	2,411	99%
Cluster 3, Rule 3	RFA_2F = 2 and E_RFA_2A = 0 and F_RFA_2A = 1	5,453	5,344	98%

Identifying Cluster Prototypes

Representative record (in natural units) of each cluster (one with closest distance to centroid)

Helpful in gaining additional insight into the meaning of each cluster.

Can examine 2nd or 3rd smallest distance as well.

FIELD	CLUSTER 1, PROTOTYPE 1	CLUSTER 2, PROTOTYPE 1	CLUSTER 3, PROTOTYPE 1
Distance	0.375	0.301	0.313
LASTDATE	9601	9601	9512
FISTDATE	9111	9109	9203
RFA_2F	3	3	1
D_RFA_2A	0	0	0
E_RFA_2A	1	1	0
F_RFA_2A	0	0	1
DOMAIN3	0	0	0
DOMAIN2	0	1	0
DOMAIN1	1	0	0
NGIFTALL	10	10	19
LASTGIFT	12	11	5

Clustering App: Feature Augmentation

↓
new feature

$$\mathbf{X}_{n \times d} = \begin{bmatrix} & \mathbf{X}_1 & \mathbf{X}_2 & \cdots & \mathbf{X}_d \\ \mathbf{x}_1 & x_{11} & x_{12} & \cdots & x_{1d} \\ \mathbf{x}_2 & x_{21} & x_{22} & \cdots & x_{2d} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mathbf{x}_n & x_{n1} & x_{n2} & \cdots & x_{nd} \end{bmatrix}$$



Clustering into
 $K \ll d$



Cluster encoding



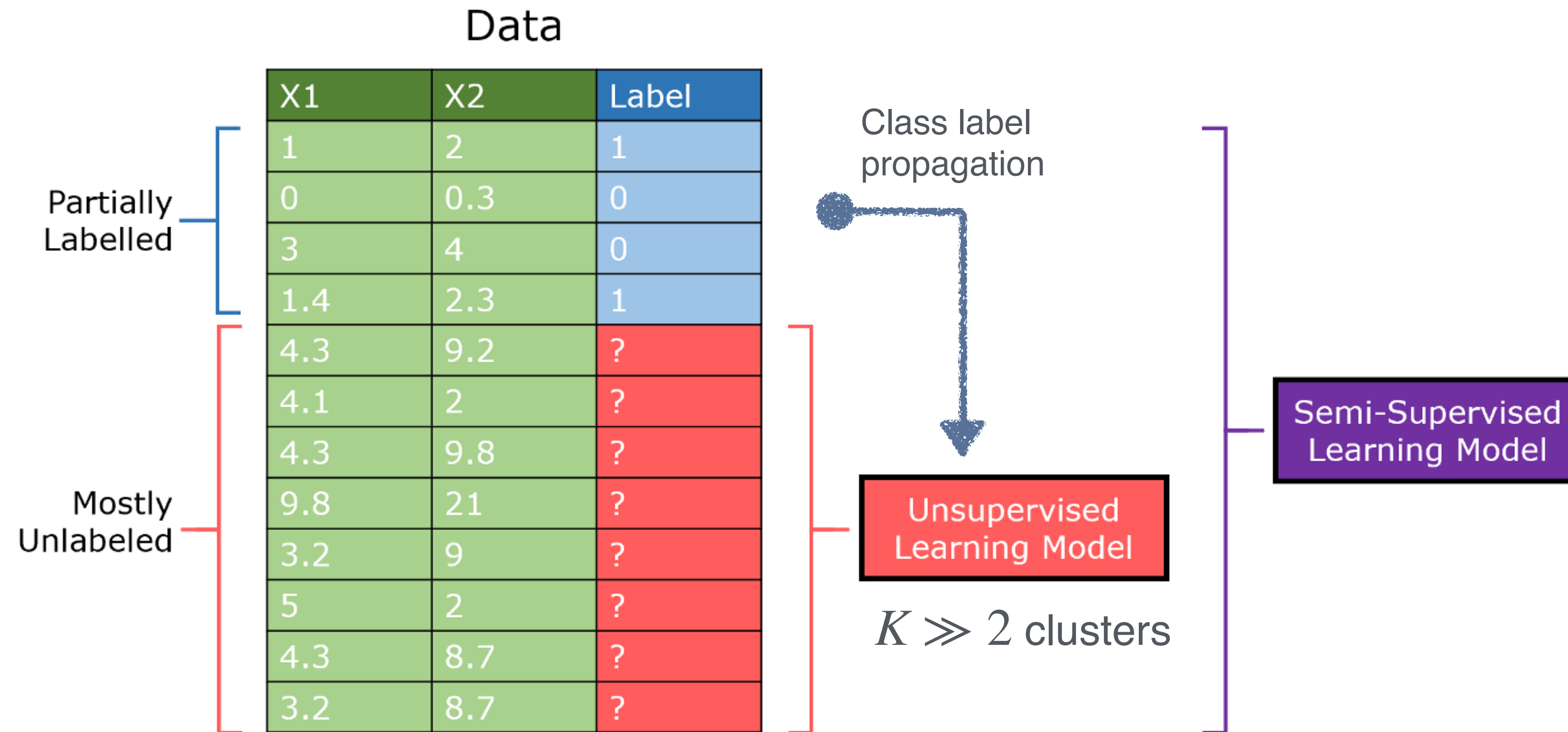
One-hot encoding
Inverse-distance encoding

Constructed features

$C_1 \ C_2 \ \cdots \ C_K$

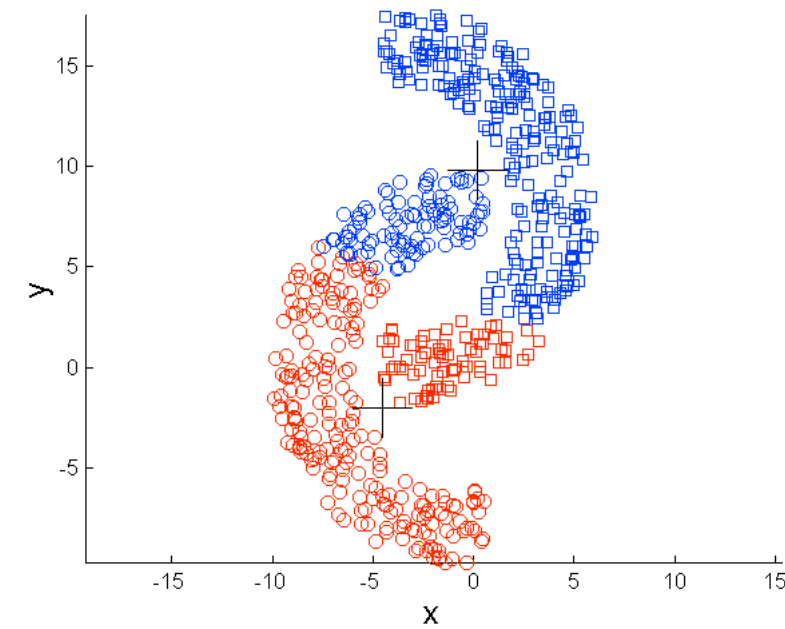
0	x		0
0	0		0
0	0		x
x	0		0
0	0		0

Clustering App: **Semi-Supervised Learning**

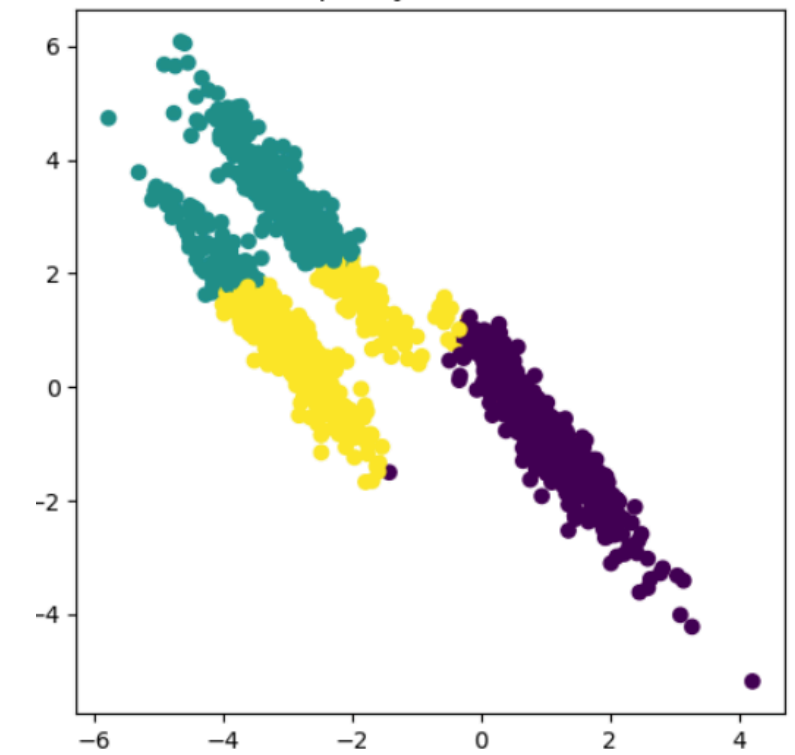


Limitations of K-Means

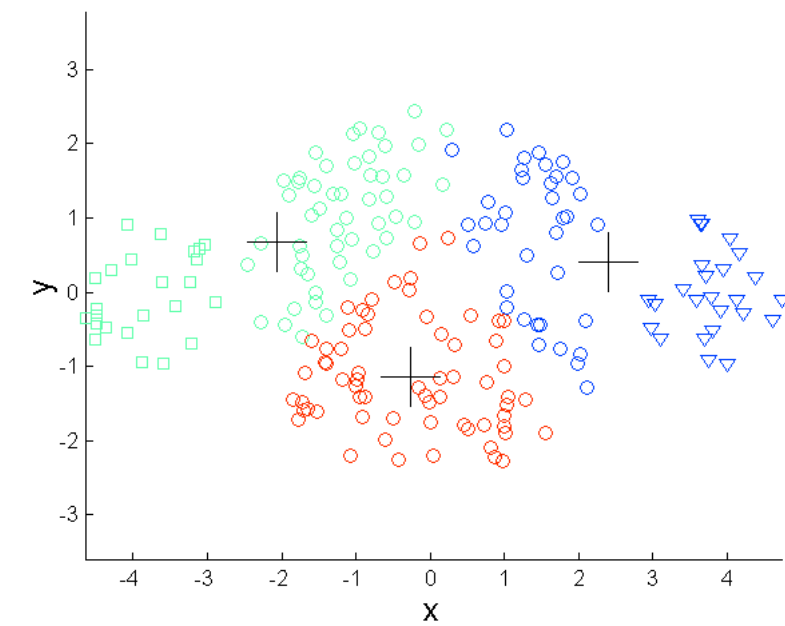
Non-convex shapes



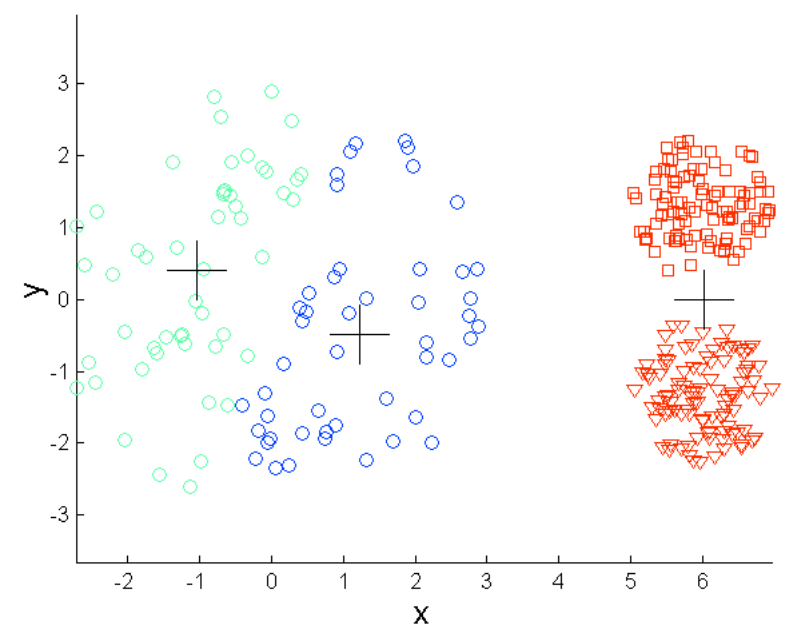
Non-spherical shapes



Different sizes



Different densities
(Unequal variances)



Summary

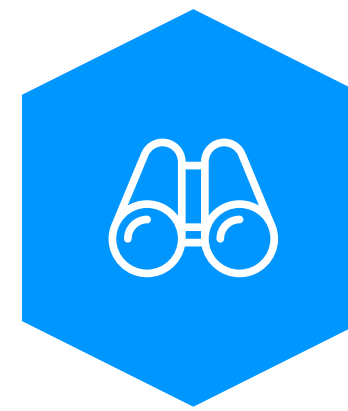
Clustering to find natural groupings of data points

Require the calculation of dissimilarity among objects.

K-means clustering

- ◆ Approximate clusterings that minimize SSE between objects and their centroids.
- ◆ Elbow method or Silhouette coefficients to determine the number of clusters

Need to interpret clustering results



Clustering Lab

(Group of 4 to 6)
